When Does a Neural Architecture Earn Its Cost?

Oruži Zarathustra Goertzel

July 2026 — working note

Abstract

Architecture choices are usually settled by benchmark. We ask a different question: under what conditions does a component *provably* earn, or fail to earn, its cost? We treat four design decisions as independent axes rather than competitors—how belief is represented, where state lives, how a coupled constraint is solved, and how parameters change—and give machine-checked boundaries for each. Fusion of independent evidence is addition in natural coordinates, and every hand-designed gating rule is that addition seen in another chart; the four ways this stops being optimal each have a closed form. Typed slots are licensed exactly when cross-slot coupling vanishes. Command routing collapses to a single gated interpolation, so it buys selectivity rather than a new update class, and order-independence of routed phases holds exactly when generators commute and an affine obstruction vanishes. Five named predictive-coding mechanisms have constructive backpropagation realizations; the residue is an execution-model property, not a training-rule advantage. Forgetting decomposes into a metric part and a connection part, and one Gram matrix certifies both a curriculum's right to be reordered and a command schedule's right to be treated as a set. Finally, in a *linear-effect* model—one that concrete linear attention and additive evidence registers provably instantiate, and that softmax provably breaks—consolidating fast state into weights is faithful exactly when the fast effect is weight-reachable; bounded capacity forces a computable consolidation deadline, and the cadence objective derived from the model's own accounting puts the optimal consolidation at that deadline, refuting the interior optimum a posited quadratic had suggested. We are explicit throughout about which results are theorems about a declared model and which quantities remain empirical; a closing section states what is assumed rather than proved, and what would discharge it. The axes are then assembled, without identifying them, in a *unified carrier*: recurrent control, typed workspace contents, belief metadata and routed operators inhabit one product state whose workspace endpoint and strict non-workspace witnesses are both machine-checked. In a sealed two-world continuation factorial over the four component carriers and five credit rules, the 1,617-entry union has large exclusive marginals along both axes; prospective PC leads the aggregate rule portfolio but not every carrier–world cell, and BP remains faster. The experiment therefore supports composition and interaction, not a universal winner. A separate dependent-proof screen then starts every credit rule at update zero: prospective PC reproducibly departs from BP and contributes held-out marginal solutions, while BP keeps higher total coverage. This motivates cost-aware composition rather than winner-take-all selection. A separate two-round fresh-start audit then shows why architecture and credit must be analyzed relationally: the Unified PC arm improves target coverage over its BP control while using fewer fresh program identities, and the advantage reverses when one high-degree reused program is removed.

Open artifacts. The formal repository, paper source, and compiled PDF are public. The source links below are immutable permalinks to commit `f3a31db36bb2f944312ee31c1d86c2758109e83a`.

1 Introduction

Debates about neural architecture are usually resolved by running two systems and comparing a number. That settles which system won a benchmark; it does not say why, and it does not transfer. This note collects a different kind of result: for each of a small number of design decisions, a condition under which the decision is provably right, and a condition—usually a fixture, often a counterexample—under which it is provably wrong.

Two commitments shape everything that follows. First, the models are exact and small: scalar and matrix linear-Gaussian systems, quadratic tasks, finite regime families. The purpose is to locate boundaries, not to predict benchmark scores, and we mark the places where a boundary is a theorem and the places where the corresponding quantity has to be measured. Second, every claim below is machine-checked in a Lean 4 development, and the negative results are checked with the same seriousness as the positive ones. Several of the sharpest statements here are refutations of things we previously believed.

The organizing idea is that the usual contest—predictive coding *versus* backpropagation, workspace *versus* recurrence, learned mixing *versus* Bayesian fusion—is a category error. These are independent axes that compose. A system chooses, separately: how accumulated belief is represented, where state lives and how it is addressed, how a coupled constraint is solved at inference time, and how parameters are updated. Asking which axis "wins" has no answer; asking when each earns its cost does. No processor designer asks whether registers beat RAM, or whether the cache-coherence protocol beats the arithmetic unit; each component answers a different question, and the architecture is the discipline of their composition. The results below are the analogous discipline for learned machines: not a tournament, but a datasheet of conditions.

2 Four axes, not four contestants

We fix the decomposition used throughout:

axis	the decision
belief coordinates	point estimate, natural coordinates, or a mixture
state carrier	recurrent control, typed slots, routed operators, or their product
inference solver	direct evaluation, or an iterative relaxation to equilibrium
plasticity rule	backpropagation, or a local energy-based update

The axes are genuinely independent: a typed-slot workspace can carry point estimates and train by backpropagation, and a single recurrent vector can carry natural coordinates and settle iteratively. The formal development reflects this—the state theorems do not mention the plasticity rule, and the plasticity theorems do not mention the state carrier.

One consequence is worth stating early, because it is easy to hope otherwise. Within the finite family of regimes we model, no single recommendation is correct everywhere: for each candidate rule there is a modelled regime in which it fails its own guarantee (`no_singleRecommendation_covers_modeledFamily`). The proof is a case analysis that exhibits, for each rule, a concrete regime defeating it. There is no universal architecture, and that is a theorem rather than a hedge.

3 A source-to-theorem programme

This note belongs to a broader, authenticated literature programme rather than a hand-picked theorem list. Its current atlas deduplicates 363 primary sources (349 arXiv records) and eight source documents. It contains 254 adjudicated primary claims and nine document claims; 254 claims are bound to checked Lean declarations. The source-level queue is deliberately visible:

source classification	sources	adjudicated claims
new formalization or stronger generalization	200	212
corresponds to an existing theorem	20	25
inapplicable: a required hypothesis fails	7	7
executable obligation, not yet a theorem about the runtime	3	6
empirical-only claim	1	4
pending primary-source audit	132	—

Presence in the corpus is not counted as coverage. A source leaves the pending row only after its statement and assumptions have been checked against the primary text and it is either connected to a theorem, reduced to an executable obligation, kept empirical, or rejected with the failed hypothesis named. A theorem entry requires recovery of the source statement, a scope boundary, and a positive and negative fixture; wrappers whose proof is merely definitional equality do not qualify.

The gradient block of [66] now has an exact compositional semantics independent of any particular neural module. A declared reverse map pairs its forward computation with a continuous-linear pullback. Inserting a stop-gradient boundary preserves the forward composite for arbitrary surrounding maps, while its upstream pullback is exactly zero (`comp_stopGradient_forward_eq`, `comp_stopGradient_pullback_eq_zero`). Replacing that boundary by reverse identity recovers the original composite, and a scalar counterfixture computes the same forward value 30 while passing a nonzero upstream credit 6 (`scalar_identity_does_not_isolate`). These results formalize declared reverse-mode semantics, not the analytic derivative of an arbitrary forward map, the InfoNCE mutual-information bound, or the paper's empirical representation claims.

The pairwise gradient criterion of [67] has a similarly sharp finite-step boundary. For a half-squared-distance retention loss, stepping along a second example's gradient changes the first loss by exactly $-\eta\langle g_1, g_2\rangle + \eta^2\|g_2\|^2/2$ (`halfSquaredDistance_gradientStep_change_exact`). Positive alignment always admits a strictly positive transferring step, while negative alignment gives strict interference at every positive step (`positive_alignment_has_safe_finite_step`, `negative_alignment_strictly_interferes`). But the sign alone is not a finite-step certificate: a scalar fixture has positive unit alignment and increases the retained loss when the rate is three (`positive_alignment_large_step_interferes`). This recovers the source's operational definition while separating it from the unformalized MER sampling and Reptile approximation.

The meta-transfer criterion of [68] supplies a complementary signal for structural change. For every positive finite categorical law, the expected score of a correctly matched unchanged mechanism vanishes along every probability-simplex tangent (`expectedScore_eq_zero_of_matched`); evaluating the same tangent under a shifted law need not vanish (`shifted_distribution_nonzero_expectedScore`). For two structural hypotheses with positive online likelihoods L_1, L_2, the exact derivative of negative log mixture likelihood is prior minus posterior and can be rewritten as

$$\frac{\sigma(\gamma)(1-\sigma(\gamma))(L_2-L_1)}{\sigma(\gamma)L_1+(1-\sigma(\gamma))L_2}.$$

Thus its sign selects the faster-adapting hypothesis (`metaTransferGradient_eq_scaled_likelihoodGap`); equal likelihoods give no structural signal, while zero likelihoods invalidate the statistical interpretation. These are score and gradient identities, not a proof that observational data identify a causal graph or that sparse mechanism shifts hold in a runtime.

Perry, von Kügelgen, and Schölkopf's sparse-mechanism-shift rate [69] supplies the missing probabilistic amplification step without hiding its causal premises. For any finite family of false candidates, independent failure events across m disjoint environment pairs make one candidate survive with probability at most r^m; a finite union bound gives

$$\Pr[\text{some false candidate survives}] \le |\mathcal{G}_{\text{false}}|\, r^m$$

without requiring independence between candidates (`measure_identificationFailure_le`). If the target mechanism changes with probability at most ρ^{UB} and a discriminating witness changes with probability at least ρ^{LB}, the one-pair failure factor is $r = 1 - (1 - \rho^{\text{UB}})\rho^{\text{LB}}$ (`pairFailureProbability_le_sparsePairFailure`). It is strictly below one under the useful nondegenerate conditions $\rho^{\text{UB}} < 1$ and $\rho^{\text{LB}} > 0$, so the finite-candidate bound tends to zero (`sparseShiftFailureBound_tendsto_zero`). Zero witness-shift probability and an always-shifted target both leave $r = 1$, making the no-decay boundary executable. The Lean result generalizes the event-probability spine of the source; faithfulness, shared mechanisms, pseudo-causal sufficiency and the argument that a wrong parent set supplies a witness remain explicit premises rather than conclusions of this theorem.

Activation-subspace projection provides a stronger exact invariant for a single linear layer. The gradient-projection memory of Saha, Garg, and Roy [3] stores orthonormal activation bases; its residual-projected fully connected update annihilates every stored basis direction exactly (`fullyConnectedUpdate_annihilates_basis`). Lin et al.'s TRGP construction [4] adds a task-indexed scaling view of that frozen subspace. Identity scaling recovers the shared weight, stored-coordinate inputs are transformed exactly by the small scaling matrix, and inputs orthogonal to the memory are unchanged (`effectiveWeight_identityScale`, `effectiveWeight_mulVec_storedCoordinates`, `effectiveWeight_mulVec_eq_weight_of_orthogonal`). Meanwhile an arbitrary residual-projected shared-weight step preserves the old projected weight for every step size (`projector_weight_preserved_after_projectedUpdate`). The task-indexing and SVD premise are essential: a nonidentity scale changes the new-task view even though the shared weight is untouched, and a nonorthonormal stored basis breaks the intended coordinate action (`fourfoldScale_view_separates_from_shared`, `nonorthonormal_memory_breaks_scaledCoordinate_action`). These are exact layerwise identities; the projected-gradient norm as a task-correlation surrogate, top-K selection, finite nonlinear retention, and empirical transfer improvements remain outside the theorem.

Equal function classes still need not have equal continual-update dynamics. Claim C.1 of Mirzadeh et al. [5] compares a direct linear model with a two-layer linear factorization. For orthogonal old and new task rows, every finite sequence of new-task gradient packets preserves the direct model's old predictions (`directPrediction_runDirectPackets_eq`), and the analogous factorized path preserves the old data-to-hidden map (`oldFeatureMap_runFeaturePackets_eq`). Nevertheless, output-head motion produces exactly one half of the squared old-feature-map image of the head displacement (`factorizedForgetting_runFeaturePackets_eq_halfSquared_headDrift`). Column independence makes every nonzero head displacement strictly forgetting (`factorizedForgetting_pos_of_columnIndependent`), whereas an executable noninjective fixture changes the head without changing the old prediction (`noninjective_featureMap_head_change_zero_forgetting`). A one-hidden-unit lift proves that this separation is dynamical rather than expressive (`collapsedWeight_oneHiddenLift`). It therefore does not turn the paper's empirical width trend into a universal nonlinear theorem.

Guha and Lakshman [6] instead give a conditional nonlinear width envelope: after collecting the task, depth, sparsity, activation and norm terms into a positive coefficient independent of the widths being compared, the remaining factor is $W^{-\beta}$ for a positive data-dependent exponent. The formalized algebra proves the exact scaling identity and the stronger diminishing-gain statement: multiplying width repeatedly by the same $s > 1$ keeps reducing the envelope, but each absolute reduction is $s^{-\beta}$ times the preceding one (`widthEnvelope_scale`, `scalingGain_strictly_diminishes`). It also exposes the capacity decision boundary. If the collected coefficient grows by a factor ℓ while width grows by s, the new envelope is smaller exactly when $\ell < s^{\beta}$ (`load_width_tradeoff_lt_iff`). Zero exponent gives no width gain and a negative exponent reverses the monotonicity (`zero_exponent_has_no_width_gain`, `widthEnvelope_strictly_increases_of_exponent_neg`). This theorem does not reprove the source's random active-row model, lazy-training assumption, spectral perturbation estimates, big-O constant, fitted exponent, or independence of the collected coefficient from width. Those are runtime and source-model premises, so width alone cannot license promotion of a larger carrier.

The continual-flow analysis of Graldi et al. [7] supplies the complementary mechanism. If the old-task residual follows the negative cross-task-kernel transport of the current new-task residual, its half-squared energy derivative is exactly the negative pairing of those two quantities (`oldTaskEnergy_hasDerivAt_crossTaskForgettingRate`). The pairing sign is therefore the exact instantaneous boundary: a negative pairing gives positive forgetting, a positive pairing gives transfer, and orthogonality gives zero rate (`crossTaskForgettingRate_pos_iff`, `crossTaskForgettingRate_eq_zero_of_inner_eq_zero`). Its absolute magnitude is bounded by the product of old-residual norm, cross-kernel operator norm and new-residual norm (`abs_crossTaskForgettingRate_le`). Positive semidefiniteness of the complete task kernel is not enough: a unit-diagonal two-task kernel with off-diagonal coupling $1/2$ is positive semidefinite for every error pair, yet equal-norm aligned and opposite residuals give rates $-1/2$ and $+1/2$, respectively (`positiveSemidefinite_jointKernel_allows_both_rate_signs`). This recovers the source differential identity but not its dynamical mean-field limit, Gaussian-process construction, self-consistent kernel system, finite-width approximation, lazy–rich transition or empirical critical feature-learning value. The registered generations-11–13 trace measures teacher-forced and fixed-budget behavioural retention, not the cross-task pairing or its finite-step remainder, so it cannot turn this identity into a retention certificate. A future retention experiment must measure both quantities explicitly.

The diagonal "Fisher" used by continual-learning code also has a load-bearing estimator boundary. Following van de Ven [8], the exact per-input contribution is the model-distribution expectation of the squared score. Drawing one class from that same distribution is unbiased, both per input and after a finite dataset average (`expected_model_sample_eq_exact`, `expectedSampledDatasetFisher_eq_exact`). This does not identify the observed-label empirical contribution with the exact Fisher. A valid three-class score model has exact contribution $1/2$ while its observed class contributes zero, and uniform rather than model-distributed label sampling has expectation $2/3$ (`empirical_zero_exact_positive`, `uniform_sampling_differs_from_exact`). Minibatch aggregation introduces a second, exact discrepancy: squaring the sum of per-example gradients equals their individual squared mass plus every pairwise cross term (`batchSquaredMass_eq_individual_add_cross`). Consequently the two gradients $(1, -1)$ cancel to zero after aggregation, whereas $(1, 1)$ are amplified from individual squared mean one to batched value four (`opposing_batch_cancels`, `aligned_batch_amplifies`). Any Fisher-weighted PC or retention profile must therefore record its label law, per-example gradients, aggregation order, and normalization before these theorems apply.

Natural continual learning (NCL) then supplies a distinct geometric use of the previous posterior precision. Equation (8) of Kao et al. [9] is recovered for every invertible symmetric positive-definite metric: applying inverse prior precision to the complete local Laplace-posterior gradient preconditions the current-task gradient while turning the quadratic restoring term into the ordinary displacement from the previous mean (`naturalContinualDirection_eq`). Completing the metric square shows that this direction is the unique minimizer of the penalized local trust-region Lagrangian and, equivalently, maximizes the linearized posterior gain over the prior-metric ball whose boundary it reaches (`trustLagrangian_sub_at_lagrangeDirection`, `lagrangeDirection_strictly_minimizes`, `lagrangeDirection_maximizes_on_own_metricBall`). Preconditioned NCL and raw Laplace-gradient updates have exactly the same fixed points at nonzero rates (`ncl_rawLaplace_same_fixed_points`), but they need not traverse the same paths: a concrete diagonal precision $\mathrm{diag}(2,1)$ maps the posterior direction $(2,1)$ to $(1,1)$, which no scalar rescaling recovers (`anisotropicNaturalDirection_eq`, `anisotropic_ncl_not_scalar_raw`). The result is local first-order trust-region geometry; it does not certify a Fisher estimator, Kronecker approximation, finite nonlinear loss decrease, or empirical retention gain.

Riemannian Walk [10] uses that metric locally along the optimizer path rather than only at the endpoint. Its online Fisher update is a convex combination that preserves $[0,1]$ under the declared coefficient bounds, while a two-event fixture proves that the result is chronological rather than permutation invariant (`movingFisher_mem_Icc`, `movingFisher_not_permutationInvariant`). For each coordinate and logged interval it divides the first-order loss decrease by a positively damped diagonal-Fisher approximation to KL displacement, accumulates the ratios, clips a negative final score to zero, and adds the result to Fisher importance in the next retention penalty. The finite scalar construction is exactly invariant under any nonzero linear coordinate rescaling when gradient, displacement and Fisher transform covariantly (`stepSensitivity_reparameterize`); replacing the Fisher metric by Euclidean displacement breaks that invariance (`euclideanSensitivity_not_reparameterizationInvariant`). The metric does not make the estimator endpoint-only: one displacement of size two and two displacements of size one reach the same parameter but produce importance $2/3$ and $4/3$, respectively (`same_endpoint_different_pathImportance`). Trace interval, actual post-optimizer displacement, damping and clipping are therefore semantic inputs. At task boundaries, repeated pairwise averaging gives three successive scores weights $1/4, 1/4, 1/2$ and is not permutation invariant (`averageImportance_three`, `averageImportance_not_permutationInvariant`). These theorems validate neither the empirical diagonal-Fisher estimator nor the finite nonlinear KL approximation; both remain runtime obligations.

Variational continual learning (VCL) [11] exposes the same order boundary at the posterior-projection level. For finite-dimensional Bayesian linear regression, the diagonal-Gaussian Sherman–Morrison mean update solves the paper's normal equation exactly, and every diagonal precision increases by its feature-square likelihood contribution (`updatedMean_satisfies_sourceEquation`, `updatedPrecision_sub_previous`). Before approximation, full Gaussian likelihood packets add in natural coordinates and therefore commute (`exactEvidence_order_independent`). Projection back to a diagonal covariance after every observation does not inherit this law. From unit prior precision and zero mean, the feature/target events $((1,0),1)$ and $((1,1),1)$ produce the same final diagonal precision $(3,2)$ in either order, but means $(3/5,1/5)$ and $(5/9,1/3)$ (`diagonalProjection_order_changes_mean`). Consequently a full-information belief ledger may fuse independent packets as a set, while a projected diagonal carrier must retain the projection schedule. Algorithm 1's coreset recursion has a second exact boundary: propagation uses $(C_{t-1} \cup D_t) \setminus C_t$, while prediction uses C_t. If $C_t \subseteq C_{t-1} \cup D_t$, these partitions are disjoint, cover all available packets, and preserve the sum of arbitrary commutative evidence weights (`coreset_schedule_exactly_once`, `coreset_schedule_preserves_evidence`). Propagating the retained coreset and then using it again for prediction counts every retained packet twice; omitting an old packet when it leaves the coreset loses it (`retained_packet_reuse_is_double_counting`, `released_old_packet_is_lost_without_transfer`). This is an exact scheduling and linear-Gaussian boundary, not a bound on repeated mean-field projection error in a neural posterior.

Bayesian Gradient Descent (BGD) [12] turns the same uncertainty coordinate into a learning-rate and curvature signal. For a scalar gradient with strong-monotonicity modulus m, each antithetic noise pair obeys

$$m\sigma\epsilon^2 \;\le\; \frac{g(\mu+\sigma\epsilon)\epsilon + g(\mu-\sigma\epsilon)(-\epsilon)}{2}.$$

The inequality sums over any finite sample family; unit empirical second moment gives the source-shaped lower bound $m\sigma$ for the average (`modulus_mul_deviation_le_averagePairedCurvature`). Equation (12)'s exact scalar deviation update stays positive and is below, equal to, or above the prior deviation according as the curvature signal is positive, zero, or negative (`uncertaintyUpdate_positive`, `uncertaintyUpdate_lt_prior_of_positive_signal`, `prior_lt_uncertaintyUpdate_of_negative_signal`). Antithetic strong curvature therefore strictly reduces uncertainty (`antithetic_strong_curvature_decreases_uncertainty`). Pairing is load bearing: one one-sided sample from a strongly convex quadratic can report a negative signal (`oneSided_sample_can_reverse_positive_curvature`). Thus a BGD-derived carrier profile needs paired-noise and second-moment telemetry; the finite theorem is not an identification with the exact Gaussian expectation or a convergence proof.

The perfect-memory argument of Knoblauch et al. [13] has a prior definitional boundary. Their displayed class $E(\theta)$ intersects all admissible solution sets containing θ; this records only

$$\theta \in A \Longrightarrow \theta' \in A.$$

It does not imply equality of membership signatures. On $\Theta = \{\bot, \top\}$ with admissible sets $\{\Theta, \{\top\}\}$, one has $\top \in E(\bot)$ but $E(\bot) \neq E(\top)$, refuting the first bullet of Lemma 3 as displayed (`source_Lemma3_first_claim_counterexample`). Defining $\theta \sim \theta'$ by $\theta \in A \leftrightarrow \theta' \in A$ for every admissible A repairs the construction: the resulting classes are equal or disjoint and every admissible set contains a whole class or none of it (`mem_membershipClass_iff_class_eq`, `membershipClass_subset_or_disjoint`). Complement closure is a sufficient missing hypothesis under which the displayed intersection equals the repaired class (`sourceIntersectionClass_eq_membershipClass_of_complementClosed`). One stored representative of every surviving repaired class then answers every admissible task-intersection query exactly (`representativeMemory_is_intersectionOracle`); omitting a class has an executable failure witness. This repairs the set-theoretic perfect-memory premise; it does not formalize the paper's NP-hardness or minimal-cardinality claims.

Gradient-based sample selection (GSS) [14] makes the replay memory question geometric. Removing remembered gradient half-spaces can only enlarge the feasible update cone, and the enlargement is strict even for two orthogonal constraints (`feasibleCone_antitone`, `constraint_reduction_strictly_enlarges`). For a fixed-size family of normalized gradients, the source's cosine surrogate satisfies

$$\sum_{i,j} \langle u_i, u_j \rangle = \left\| \sum_i u_i \right\|^2, \qquad \mathrm{Var}(u) = 1 - \frac{1}{M^2} \sum_{i,j} \langle u_i, u_j \rangle.$$

Thus minimizing the surrogate exactly maximizes directional variance (`pairwiseSimilarity_eq_norm_sum_sq`, `pairwiseSimilarity_le_iff_variance_ge`). It does not, however, identify the retained cone: the opposite pairs $\{e_1, -e_1\}$ and $\{e_2, -e_2\}$ have equal zero surrogate and unit variance, yet the first forbids direction e_1 while the second permits it (`equal_surrogate_and_variance_but_different_feasible_cones`). Accordingly, the paper's observed surrogate–solid-angle correlation remains empirical rather than a general theorem. The greedy score boundary is also exact: cosine plus one lies in $[0, 2]$, but antipodal unit gradients attain zero (`antipodal_shiftedUnitSimilarity_eq_zero`), so a normalized sampling distribution requires a separately checked positive total score.

Perfect generated inputs do not certify perfect replay targets. Deep generative replay [15] trains on current labelled data together with generated past inputs labelled by the previous solver, whereas its joint-training comparator uses true past inputs and labels. Let r be the current-data fraction, R_G the risk under generated inputs and teacher targets, R_T the risk under true past inputs and teacher targets, and R_Y the risk under true past inputs and labels. The train–test discrepancy has the exact decomposition

$$R_{\text{replay}} - R_{\text{joint}} = (1-r)\big((R_G - R_T) + (R_T - R_Y)\big) \quad (\texttt{replayTrainRisk\_sub\_jointTestRisk}).$$

Recovering the old input distribution removes the first term only. For $r < 1$, perfect generated-input replay equals true-label joint risk exactly when the teacher and true-label risks agree (`perfectGenerator_train_eq_test_iff_teacherRisk_eq`). Exact generator and teacher risks are sufficient, and the absolute train–test gap is bounded by the retained weight times the sum of their two absolute errors (`exactGenerator_exactTeacher_replay_eq_joint`, `abs_replayTrainRisk_sub_jointTestRisk_le`). A perfect-generator fixture with mismatched teacher targets fails joint training, while a separate fixture shows that the two errors may cancel (`perfectGenerator_teacherMismatch_not_joint`, `generator_teacher_error_cancellation`). Thus separate generator and teacher certificates are a robust sufficient boundary, not a necessary decomposition of every zero aggregate gap. These are risk identities; they do not establish GAN convergence, teacher calibration, optimization convergence, privacy, or benchmark retention.

Reservoir replay needs uniform eviction as well as uniform admission. Dark Experience Replay [16] maintains a fixed-capacity buffer without knowing the final stream length. If the buffer capacity is $m > 0$, $n > 0$ items have been seen, and every old item has marginal m/n, then admitting item $n+1$ with probability $m/(n+1)$ and conditionally evicting each occupied slot with probability $1/m$ gives

$$\frac{m}{n}\left(1 - \frac{m}{n+1}\frac{1}{m}\right) = \frac{m}{n+1}.$$

The incoming item has the same marginal. Lean proves this step, its propagation through every finite continuation using only the current stream length, validity of the probability ratios when $0 < m \leq n$, and exact expected occupancy (`oldMarginalAfterStep_eq_uniformMarginal_succ`, `propagatedOldMarginal_eq_uniformMarginal`, `uniformMarginal_mem_Icc`, `sum_uniformMarginals_eq_capacity`).

Uniform conditional eviction is load bearing. With capacity two after two items, admitting the third with the correct probability $2/3$ but using conditional eviction probabilities $3/4$ and $1/4$ leaves the two old marginals at $1/2$ and $5/6$, rather than $2/3$ (`biasedEviction_breaks_equalMarginals`). Zero capacity supplies no normalized occupied-slot distribution (`uniformEvictionProbability_zero`). These results certify equal marginals and expected occupancy, not the joint distribution over buffer subsets, independence of inclusion events, stored logit quality, calibration, or continual-learning performance.

Low endpoints certify a linear mode only under a shape hypothesis. MC-SGD [17] evaluates task losses on straight parameter-space paths from earlier solutions to a common candidate. If a loss is convex, the value at interpolation coordinate $t \in [0, 1]$ is at most the affine combination of its endpoint losses; hence a common endpoint budget controls the entire segment (`convexOn_univ_segmentPoint_le`, `convexOn_univ_segmentPoint_le_endpointBudget`). Without convexity, low endpoints alone say no such thing: the scalar loss $4w(1 - w)$ is zero at 0 and 1 but equals one at the midpoint (`lowEndpoints_highMidpoint`, `nonconvexBarrierLoss_not_convexOn_univ`).

The source's finite-grid path approximation also has an exact bookkeeping boundary. A grid written as zero, interior coordinates and one decomposes into the anchor loss, interior penalty and candidate loss (`discretePathLoss_zero_cons_append_one`). But minimizing their aggregate does not imply minimizing each task separately. For scalar task losses w^2 and $(w-2)^2$, every grid reduces to its sum-of-squared-coordinates times their aggregate. At positive grid weight the unique aggregate minimizer is $w = 1$, although task one is strictly better at 0 and task two at 2 (`twoTaskDiscretePathObjective_eq_weighted`, `quadraticTwoTaskPathObjective_eq_one_iff`, `aggregate_minimizer_not_component_minimizer`). This corrects the informal inference following Equation 4; it does not refute the finite objective itself. Neural-loss convexity, empirical connectivity, quadrature accuracy, optimizer convergence and benchmark gains remain empirical or executable obligations.

A task-free plateau flag is exactly once only if its two tests do not overlap. The online MAS protocol of Aljundi, Kelchtermans, and Tuytelaars [18] consolidates importance weights when the current loss-window mean and standard deviation fall below their plateau thresholds. Its flag then blocks further consolidation until a peak satisfies

$$\mu_{\text{old}} + \sigma_{\text{old}} < \mu_{\text{window}}.$$

Lean models Algorithm 1's two tests in their printed sequential order. A consolidation with no simultaneous peak disarms the gate, so an unchanged window cannot emit again; a later peak rearms it (`same_observation_does_not_repeat_of_not_peak`, `step_armed_of_peak`). The sufficient certificate

$$\delta_\mu \le \mu_{\text{old}} + \sigma_{\text{old}}$$

makes the plateau and peak mean inequalities disjoint and yields an exactly-once unchanged-window theorem (`plateau_not_peak_of_meanThreshold_le_peakBoundary`, `same_observation_does_not_repeat_of_threshold_order`).

This condition is not syntactic bookkeeping. Because the source prints two independent `if` statements, an observation satisfying both tests first consolidates and then immediately rearms the flag. The same observation can therefore consolidate again (`simultaneous_peak_allows_same_observation_repeat`, `sequential_if_overlap_repeats_consolidation`). A separate plateau–peak–plateau fixture realizes the intended lifecycle. The state-machine result does not establish that low loss statistics identify a semantic task plateau, that a hard-example buffer is representative, that importance estimates are calibrated, or that the method retains accuracy.

The closed-form pure-replay model of Banayeeanzade, Soltanolkotabi, and Rostami [19] makes the memory budget itself nonmonotone. Write $D = \|w_1^\star - w_2^\star\|^2$, $A = \|w_1^\star\|^2$, and $s = p - n - m - 1$. Equation (18) is

$$F(m) = \frac{n}{p}\left(1 + \frac{m}{s}\right) D - \frac{m}{p} A.$$

Its exact one-sample increment is

$$F(m+1) - F(m) = \frac{n(p-n-1)D - A\,s(s-1)}{p\,s(s-1)} \quad (\texttt{pureReplayForgettingClosedForm\_memoryStep\_difference}).$$

Hence, when $p > 0$ and $s > 1$, one more replay sample strictly improves the expression exactly when $n(p-n-1)D < A\,s(s-1)$, and strictly worsens it under the reverse inequality (`pureReplayForgettingClosedForm_add_memory_lt_iff`, `pureReplayForgettingClosedForm_lt_add_memory_iff`). The boundary is not cosmetic: at $(p, n, m, A) = (10, 2, 1, 1)$, increasing memory from one to two moves F from $2/15$ to $2/25$ when $D = 1$, but from $3/5$ to $16/25$ when $D = 3$. This is a derived theorem about the source's two-task noiseless Gaussian linear closed form. It does not reprove the random-matrix expectation, transfer the formula to nonlinear carriers, or license replay or persistent evidence without measuring the required quantities.

Fixed context-dependent gating (XdG) [20] makes a complementary claim about where a task may write. For active endpoint sets A_{out} and A_{in}, the active connection support is their Cartesian product, hence

$$|S| = |A_{\text{out}}|\,|A_{\text{in}}|, \qquad |S_1 \cap S_2| = |A_{\text{out},1} \cap A_{\text{out},2}|\,|A_{\text{in},1} \cap A_{\text{in},2}|.$$

These identities are exact (`card_connectionSupport`, `card_connectionSupport_inter`). If a new task's support is disjoint from an old task's support, its masked gradient update preserves the old masked linear readout (`disjoint_support_update_preserves_maskedLinear`). The source's concrete 400-active-of-2,000 construction consequently activates 160,000 of four million hidden-to-hidden connections—exactly four percent—and leaves ninety-six percent inactive. The layer-local scope is load-bearing: disjoint hidden supports do not protect a shared downstream bias (`disjoint_hidden_support_does_not_protect_shared_bias`). Thus task masks can certify a local noninterference boundary, but the source's whole-network retention and its complementarity with SI or EWC remain empirical until shared heads, optimizer state, normalization state, and every mutable path are bound.

Latent replay [21] exposes the corresponding representation-state boundary. Let f be the representation below the replay site. If every cached activation equals the current $f(x)$ for its raw example, then a replay step from cached activations is exactly the native-rehearsal step (`latentReplayStep_eq_nativeReplayStep_of_cacheMatches`). For a finite representation trace $z_0, \ldots, z_T$,

$$\|z_T - z_0\| \le \sum_{t<T} \|z_{t+1} - z_t\|,$$

and an L-Lipschitz downstream head transports this into an output mismatch bounded by L times the same path variation (`norm_finalLatent_sub_initial_le_latentPathVariation`, `norm_head_final_sub_initial_le_pathVariation`). A constant representation path has zero variation, while refreshing a cache at the current representation resets its pointwise age to zero. Crucially, "frozen weights" is weaker than a frozen representation map: a scalar stateful-normalization fixture keeps the weight fixed but changes its running state, thereby aging the cached activation and changing the downstream output (`fixed_weight_mutable_normalization_ages_cache`, `mutable_normalization_changes_downstream_output`). Consequently, latent or frozen-cone replay is exact only after parameters, normalization buffers, mode, randomness, preprocessing, and every other representation-affecting state have been authenticated; the source's storage, run-time, accuracy, and refresh-rate findings remain empirical.

Compute-budgeted adaptive layer freezing [22] supplies an explicit exchange criterion for deciding when such freezing is worthwhile. If R is the anticipated information-per-compute rate, b_n the backward cost of layer n, q_n its current-batch Fisher-trace proxy, and B_n the criterion for freezing the first n layers, then

$$B_{n+1} = B_n + (Rb_n - q_n).$$

Hence extending a frozen prefix is weakly beneficial exactly when $q_n \le Rb_n$, and strictly beneficial exactly when the inequality is strict (`batchFreezingCriterion_le_succ_iff`, `batchFreezingCriterion_lt_succ_iff`). The layerwise condition lifts to a nonnegative certificate for an entire prefix. It cannot be replaced by a global ratio: a two-layer fixture makes freezing the first layer the unique best global Fisher-per-compute choice while giving it a negative criterion on a shifted current batch (`twoLayer_global_rate_prefers_freeze_first`, `global_best_prefix_can_have_negative_batch_criterion`).

The paper's similarity-aware replay distribution has a second exact boundary. At positive temperature it is normalized and strictly prefers lower effective-use scores (`sum_retrievalProbability_eq_one`, `retrievalProbability_gt_of_effectiveUse_lt`). At zero temperature exact-real division maps all logits to zero and erases this order. Moreover, cosine-valid negative class similarity can make effective use negative despite nonnegative use counts (`negative_similarity_can_make_effectiveUseFrequency_negative`). Thus it is a signed routing score unless stronger hypotheses are checked, not a literal frequency. Fisher trace remains a proxy: future knowledge, reinvestment of saved work, estimator fidelity, FLOP accounting, and accuracy remain executable or empirical obligations.

Matched update counts are not matched computation. The computationally budgeted protocol of Prabhu et al. [23] charges auxiliary teacher and selection forwards in the same work unit as ordinary training. If one baseline update costs two half-baseline units, distillation costs three and costly sampling costs four. The source's ratios are then exact on divisible budgets:

$$W_{\mathrm{base}}(3k) = W_{\mathrm{distill}}(2k), \qquad W_{\mathrm{base}}(2k) = W_{\mathrm{sample}}(k)$$

(`distillation_two_thirds_exact`, `costlySampling_one_half_exact`). For every positive per-update cost, integer division gives the greatest affordable update count (`affordableIterations_maximal`). The rounding boundary is load-bearing: at a nominal baseline budget of one hundred updates, sixty-six distillation updates cost 198 half-units, the baseline costs 200, and rounding upward to sixty-seven costs 201 (`distillation_rounding_boundary_at_one_hundred`).

The same finite arithmetic governs a fixed horizon split across stream steps. All work is allocated exactly iff the number of steps divides the total budget (`allocatedAcrossSteps_eq_iff_dvd`); the source's 8,000-update horizon divides into 400, 160, and 40 updates across 20, 50, and 200 steps, respectively. A zero-cost profile has no meaningful finite affordability maximum, and a nondivisible horizon leaves a remainder. These are accounting theorems only: traces must still bind a common work unit and measure auxiliary operations, elapsed GPU work, and completed updates, while accuracy and verified discovery yield remain empirical.

A real-time stream does not wait for a slow learner. The protocol of Ghunaim et al. [25] treats relative complexity as a schedule, not merely a statistic. If a method update consumes $p > 0$ work units and each stream arrival supplies $q > 0$, then after T arrivals the greatest affordable whole-update count is

$$U(T) = \left\lfloor \frac{Tq}{p} \right\rfloor .$$

The formal scheduler proves maximality, exact conservation of consumed plus residual work, and a residual below one method update (`completedUpdates_maximal`, `consumed_add_residual_eq_available`, `residualWork_lt_methodWork`). If $q \leq p$, then $U(T) \leq T$ and the remaining arrivals are precisely the skipped stream steps.

Rational rates are load-bearing. On a common 60-step horizon, relative complexities $1, 4/3, 2, 5/2, 3, 6$ complete respectively $60, 45, 30, 24, 20, 10$ updates (`source_table_sixty_step_schedule`). Rounding $4/3$ down to the integer delay one incorrectly grants 60 updates rather than 45 (`rounding_four_thirds_to_one_overallocates`). Accordingly, future PC comparisons must record the model version used for every prediction, selected and skipped batches, work units, wall/GPU time, and checker parity. The arithmetic does not validate FLOP proxies or establish accuracy or discovery yield.

Exact memory consistency is conditional. Adaptive Continual Memory [26] combines a fixed feature extractor with approximate nearest-neighbor storage. Its exact finite core can be stated without an optimizer: for the discrete metric, exact one-nearest-neighbor lookup with an explicit most-recent-first tie rule recalls a newly prepended example immediately (`inserted_example_is_immediately_recalled`). Every finite prefix of later inserts whose features differ from the query preserves that stored label (`distinct_future_inserts_preserve_stored_example`).

The hypotheses are load-bearing. A conflicting duplicate feature can reverse the old prediction (`conflicting_duplicate_can_reverse_old_prediction`); with $k = 3$, two nonmatching labels can outvote a newly inserted exact match (`three_neighbor_majority_can_defeat_exact_insert`); and changing only the encoder can reverse recall while leaving memory byte-for-byte unchanged (`encoder_drift_can_reverse_recall`). Thus a zero optimization stability gap does not by itself imply prediction invariance. Approximate HNSW recall, logarithmic retrieval cost, wall time, and accuracy remain empirical obligations rather than consequences of the exact-memory theorem.

Inverse-frequency replay needs capacity reconciliation. Frequency-Aware Replay [24] gives each tracked class a weight inverse to the number of stream experiences in which it has appeared. For a finite nonempty class set with positive observation counts, the normalized weights are positive, sum to one, and strictly favor a less frequently observed class (`sum_normalizedInverseQuota_eq_one`, `normalizedInverseQuota_gt_of_observations_lt`). If a buffer of capacity M independently rounds every real-valued class request upward, the total request S satisfies the sharp general envelope

$$M \leq S < M + |\mathcal{C}|$$

(`capacity_le_totalRoundedSlots`, `totalRoundedSlots_lt_capacity_add_card`).

The lower inequality need not be equality. Two classes observed once and twice receive normalized quotas $2/3$ and $1/3$; a one-slot buffer therefore requests $\lceil 2/3 \rceil + \lceil 1/3 \rceil = 2$ slots (`two_class_ceil_oversubscribes_one_slot`). Moreover, totalized rational division gives a zero-count class zero inverse weight, rather than infinite priority (`zero_observation_is_not_prioritized`), so initialization at one is essential. Any use in the unified carrier must consequently trace the positive-count invariant and an explicit post-rounding allocation rule; accuracy and forgetting reduction remain empirical.

The printed ordered-pair task-confusion identity double-counts. Nori and Kim [41] decompose a multiclass discriminative loss into pairwise class losses. Equation (2) as printed sums every ordered distinct pair while using the normalization for an unordered pair sum. If ℓ_i is the additive contribution of class i, the exact finite identity is

$$\sum_i \sum_{j \neq i} (\ell_i + \ell_j) = 2(N-1) \sum_i \ell_i$$

(`orderedPairLoss_eq`). Thus the printed factor $1/(N-1)$ returns twice the original loss; $1/(2(N-1))$ repairs the ordered-pair formula (`printed_ordered_normalization_eq_twice`, `corrected_ordered_normalization`). Already for two class contributions 3 and 5, the displayed ordered expression evaluates to 16 while the original loss is 8 (`printed_two_class`). This refutes the equality as printed, but a positive constant rescaling preserves minimizers, so it does not by itself refute every downstream optimization argument.

The source's derivative incompatibility core does survive: a nonzero cross derivative at an individual differentiable minimum excludes that point from minimizing the summed objective (`crossDerivative_excludes_sum_minimizer`). Its generative feasibility statement is conditional in a second, load-bearing way. If inter-task confusion is identically zero and one parameter simultaneously minimizes every diagonal block, then it minimizes the total objective (`zero_confusion_and_simultaneous_minima_are_globally_optimal`). Block diagonality alone does not supply the shared parameter: the scalar blocks x^2 and $(x-1)^2$ have unique, incompatible minimizers (`block_diagonal_does_not_imply_simultaneous_minimizer`). Accordingly, the theorem identifies what follows from a common minimizer; it does not prove that such a minimizer exists for a generative learner.

The utility statistics of continual backpropagation [73] reveal a source-level indexing boundary. An exponential average initialized at zero and fed a constant signal is exactly $(1 - \eta^a)$ times that signal after age a (`constantEmaRun_eq`); dividing the current accumulator by this factor therefore recovers the signal. Equation (4) as typeset instead uses the previous accumulator. Its literal finite semantics returns zero at age one and $2/3$, rather than one, for decay $1/2$ at age two (`printedLaggedBiasCorrection_age_one`, `printedLaggedBiasCorrection_half_age_two`). The replacement mechanism has a separate exact invariant: transferring mean-activation times outgoing weight to the consumer bias before zeroing that weight preserves preactivation at the mean, with off-mean drift exactly equal to weight times mean-minus-activation (`transferMean_zeroWeight_preserves_at_mean`, `transferMean_zeroWeight_change_exact`). These are finite mechanism theorems, not a proof of the paper's empirical plasticity claims.

The last-layer "zapping" mechanism of Frati et al. [74] has a complementary upstream-credit boundary. For a finite linear head and half-squared loss, the representation gradient is the prediction residual times the head vector, and a finite feature perturbation obeys an exact quadratic expansion (`halfSquaredFeatureLoss_add_exact`). Consequently a zero head sends no upstream gradient, opposite heads reverse the gradient at a zero feature, and two heads can make the same present prediction while inducing different representation gradients (`equal_prediction_different_featureGradient`). On the frozen scalar unit-feature restriction, however, T head steps leave the residual equal to $(1-\eta)^T$ times its initial value (`runUnitFeatureHeadSteps_sub_target`). The same nondegenerate schedule therefore preserves the strict ordering of two initial losses (`equalSchedule_loss_lt_iff`). A reset that starts farther from the fixed-feature target cannot become beneficial through identical head relearning alone; the source's benefit must use representation training, path dependence, stochastic resampling, or another mechanism absent from this restriction. The result neither proves nor refutes the reported transfer improvements.

Linear CKA supplies a caution about using representation similarity to judge these mechanisms. In the finite squared formulation, independent nonzero isotropic rescalings leave the statistic unchanged (`linearCkaSquared_scale`), and every dot-product-preserving feature map leaves the entire Gram signature unchanged (`linearCkaSquared_map_of_dotPreserving`). This recovers and slightly strengthens the orthogonal and positive-scaling invariances of Kornblith et al. [75]. The boundary is constructive: the invertible feature map $(x_1, x_2) \mapsto (x_1, 2x_2)$ changes a centered fixture's squared CKA from 1 to 125/152 (`invertibleAnisotropicRescaling_changesCka`). Davari et al.'s subset-translation mechanism [76] has a separate function-preserving witness. Translating one of three samples in a feature ignored by the readout and then recentering preserves that readout on every sample (`subsetTranslation_preserves_readout`), while squared CKA with the original representation falls to 5/8 (`functionPreserving_subsetTranslation_changesCka`). Thus neither a high nor a changed CKA value alone establishes functional or mechanistic equivalence. The finite witness does not claim the source's asymptotic subset-translation formula or its empirical intervention results.

Cyclical learning rates provide a complementary finite-time example in which one step cannot be judged in isolation. On a scalar quadratic mode, an arbitrary finite rate schedule acts by the product of its per-step multipliers (`runRateSchedule_eq_scheduleGain_mul`); repeated cycles raise that gain to the cycle count (`runRateCycles_eq_scheduleGain_pow_mul`). This yields an exact instance of Smith's temporary-harm intuition [77]: a unit-curvature rate-3 step quadruples loss, but following it by rate 3/4 makes the two-step cycle quarter the starting loss and beat two matched rate-1/4 steps (`largeThenCorrectiveCycle_beats_twoQuarterRateSteps`). The nearby cycle $(3, 1/4)$ instead has gain $-3/2$ and increases loss (`unstableLargeCycle_increases_loss`), so a large maximum rate alone does not license the one-cycle mechanism of Smith and Topin [78]. Moreover, fixed scalar curvature makes the gain invariant under schedule reversal (`scheduleGain_reverse`). Order-specific gains and the reported super-convergence therefore require curvature variation, noncommuting modes, stochasticity, momentum, regularization, or empirical structure outside this theorem.

The missing curvature-feedback mechanism has its own exact reduced boundary. Cohen et al.'s scalar quadratic calculation [79] is recovered inside the declared two-coordinate edge-of-stability map: the unstable coordinate is multiplied by $1 - \eta(2/\eta + y) = -(1 + \eta y)$ (`step_unstable_eq_quadratic_multiplier`). Thus positive sharpness offset strictly grows every nonzero unstable amplitude, while $-2 < \eta y < 0$ strictly contracts it (`unstableAmplitude_grows_above_edge`, `unstableAmplitude_contracts_below_edge`). Declaring equation (3) of Damian, Nichani, and Lee [80] as an exact finite map makes its negative feedback algebraically sharp: the offset decreases iff $2\alpha < \beta x^2$, increases iff $\beta x^2 < 2\alpha$, and is preserved exactly at equality (`sharpnessOffset_decreases_iff`, `sharpnessOffset_increases_iff`, `sharpnessOffset_` preserved_iff). At zero offset the balance surface has a sign-flipping period-two orbit (boundary_twoCycle); conversely, zero unstable amplitude removes the stabilizing term and leaves an arithmetic sharpening trajectory (iterate_zero_unstable). These are exact theorems of the reduced map, not claims that arbitrary neural losses satisfy the source's Taylor, eigengap, remainder, ODE-tracking, or empirical hypotheses.

The one-unstable-mode central flow of Cohen et al. [81] sharpens this boundary into an exact projection identity. In the declared reduced flow, the unique variance cancelling the affine sharpness drift is $2\langle \nabla S, -\nabla L\rangle / \|\nabla S\|^2$ (sharpnessDrift_eq_zero_iff, oneModeVariance_balances_sharpness). Substituting it into the sharpness-penalized direction gives exactly the negative loss gradient with its sharpness-gradient component removed (penalizedFlow_eq_centralFlow). The resulting direction is orthogonal to ∇S and has nonpositive first-order loss derivative at every nonnegative learning rate (inner_sharpnessGradient_centralFlow_eq_zero, inner_lossGradient_centralFlow_nonpos). This projection is the same Hilbert-space least-squares residual used by the depth-one Anderson theory, rather than a second ad hoc construction. Its limits are also exact: a loss gradient parallel to a nonzero sharpness gradient leaves no admissible direction, while regressive sharpening makes the formula request a negative variance (centralFlow_eq_zero_of_lossGradient_eq_smul, regressive_negative_variance). The paper itself calls the time-averaging derivation informal; these theorems certify equations (12)–(15) of the reduced model, not discrete neural-trajectory tracking, Taylor remainders, eigenspace assumptions, or the empirical predictions.

Adaptive restart supplies a finite safeguard for accelerated blocks without turning a heuristic into a universal convergence theorem. The function scheme of O'Donoghue and Candès [82] is exactly rejection by the zero-tolerance endpoint gate (functionRestart_iff_endpointRejected); the gradient scheme is exactly the positive inner-product test, equivalently an obtuse angle between momentum displacement and negative gradient (gradientRestart_iff_negativeGradient_obtuse). When the gradient and objective comparison share an anchor, a first-order lower bound makes the gradient test imply a function restart (gradientRestart_implies_functionRestart_of_sameAnchor). The anchor condition is substantive: one scalar-quadratic fixture triggers the lookahead-gradient restart while the objective improves, while another raises the objective without triggering the same-anchor gradient test (lookaheadGradient_can_restart_during_improvement, functionIncrease_without_sameAnchorGradientRestart). Operationally, restart preserves the current position, zeros the extrapolation displacement, and is idempotent (momentumDisplacement_resetMomentum, resetMomentum_idempotent). A concrete settling implementation must still establish the source's convex or locally quadratic hypotheses before claiming its accelerated rate.

Checkpoint averaging has a similarly exact admission boundary. Section 3.5 of Izmailov et al. [83] compares the prediction at an averaged weight with the average of predictions by centering the sampled weight displacements. The formal result generalizes uniform scalar averaging to arbitrary nonnegative affine weights and real normed parameter and output spaces: affine normalization removes the constant term, centered displacements remove the complete linear term, and a declared pointwise Taylor-remainder budget controls the remaining aggregate error (weightedPrediction_sub_center_eq_remainder, norm_weightedPrediction_sub_center_le). Affine predictors commute with averaging exactly (weightedPrediction_affine_eq_center). Both premises are sharp: centered weights under a scalar-square predictor have a unit second-order gap, while noncentered weights fail even for the identity predictor (centered_square_prediction_has_unit_gap, noncentered_affine_prediction_mismatch). Consequently weight averaging is a theorem-licensed approximation only after a trace supplies centering and local remainder evidence; the source's wider-optimum, generalization, batch-normalization and accuracy results remain empirical.

One recent tranche illustrates the difference between bibliography and formalization. The PReLU recurrence of He et al. is recovered as an exact finite-depth second-moment product;

the prescribed scale preserves the moment, while zero fan-in and using the ReLU scale at nonzero PReLU slope are proved failure boundaries. The residual equations of He et al. are generalized to a heterogeneous product transport bound, with identity at zero branch and expansion at the identity branch. Fixup's scalar branch condition is strengthened to heterogeneous preceding coefficients, separating a live zero-output last-layer update from a dead zero prefix. The Pre-LN/Post-LN ordering of Xiong et al. yields an exact update-zero separation and a two-coordinate counterexample: a zero Pre-LN branch is identity, while Post-LN still changes a non-fixed state.

The correlation linearization of [59] is also recovered as an exact finite scalar transport law. For every positive subcritical multiplier $\chi < 1$, the residual after L layers is $\chi^L \varepsilon_0 = \exp(-L/\xi_c)\varepsilon_0$ with $\xi_c^{-1} = -\log\chi$ (`linearizedResidual_eq_depthScale_exp`). A requested finite propagation horizon lies below ξ_c exactly when $\exp(-1/H) < \chi$, and a constructive theorem supplies such a strictly subcritical multiplier for every $H > 0$ (`budget_lt_correlationDepthScale_iff`, `exists_subcritical_multiplier_with_depthScale_gt`). The ordered, critical and chaotic cases respectively contract, preserve and expand every nonzero linearized residual. Equation (13)'s independent-mask dropout term is likewise exact at correlation one: nontrivial dropout lowers correlation under positive variance, but an identically zero activation moment leaves it fixed (`dropoutCorrelationAtOne_lt_one`, `dropoutCorrelationAtOne_zero_activationMoment`). These are conditional algebraic gates for measured runtime quantities, not a formalization of the Gaussian mean-field approximation or a guarantee of trainability.

Equation (8) of [60] supplies a coupled geometric recurrence. Taking its Euclidean metric and squared-curvature updates as finite dynamics, the metric is exactly $\chi_1^L g_0$, while curvature error around the declared fixed point is exactly $(1/\chi_1)^L$ times its initial error (`euclideanMetric_eq_pow_mul`, `curvatureSq_sub_fixedPoint_eq`). More revealingly, squared curvature times Euclidean metric obeys an exact product law: the inherited curvature cancels the metric stretch, leaving precisely the new nonlinear curvature-injection term (`gaussMetric_succ`). Positive injection therefore gives strict layerwise growth, whereas zero injection conserves the product even when the linear metric expands (`gaussMetric_succ_strictly_increases`, `gaussMetric_zero_curvatureSource`). This formalizes the finite recurrence and its boundary, not the random-network mean-field derivation, activation integrals, manifold limit, shallow lower bound or simulations.

The spatial hierarchy in [61] adds a second boundary. Abstracting a Fourier mode to any finite family of unit complex phases, every nonnegative variance kernel of total mass one has mode-gain norm at most one (`norm_modeGain_le_one`) and preserves the constant mode exactly. A point-mass kernel gives every phase unit gain, hence preserves every such mode at criticality (`critical_pointKernel_modeResidual`). By contrast, the normalized two-site uniform kernel preserves the constant mode but annihilates the alternating mode after one layer (`uniformTwoKernel_separates_modes`, `uniformTwoKernel_alternating_annihilated`). Thus average criticality cannot certify preservation of spatial structure. This finite mode theorem does not recover the paper's random-channel limit, multi-channel orthogonal-kernel construction or end-to-end Jacobian spectrum; those remain separate obligations.

The one-hidden-layer linear argument of [62] gives a complementary width boundary. Expanding one simultaneous update of $f(x) = V^\top Ux$ yields four exact dot-product terms (`twoLayerLinearOutput_after_update`). Under orthogonal standard statistics, its output is strictly increasing with width whenever the input–signal product is positive (`standardOneStepOutput_strictMono_width`). Under the source's maximal-update squared-norm statistics and reciprocal input/output-layer rates, any two positive-width finite networks instead have exactly the same one-step output (`maximal_update_statistics_width_invariant`). Rational width-four witnesses begin at the same zero output and move to five under standard scaling versus two under maximal-update scaling. This theorem recovers the paper's finite algebraic mechanism, not its random or infinite-width limit, optimizer generalization, optimal-hyperparameter claim, or empirical zero-shot transfer.

The depthwise residual prescription of [63] exposes why a scalar coefficient is only meaningful together with branch geometry. For orthonormal residual directions, the squared norm of the accumulated effect is exactly the sum of squared coefficients (`orthonormal_residualEffect_norm_sq_eq`), so the uniform $1/\sqrt{L}$ scale has depth-independent quadratic budget (`sqrtDepthScale_quadraticBudget`). Fully aligned unit directions give the opposite boundary: the same scale has squared effect proportional to L, whereas uniform $1/L$ preserves their total effect but makes the orthogonal quadratic budget decay. No uniform scalar coefficient can make both budgets one once $L > 1$ (`no_scalar_depthScale_preserves_both_geometries`). Consequently any promoted depth profile must record cross-branch geometry as well as its coefficient. These finite identities neither establish the paper's dynamical mean-field limit nor its empirical hyperparameter transfer.

[64] sharpens that one-coefficient law into a paired branch-and-update prescription. If the residual multiplier and the effective parameter update both scale as $1/\sqrt{L}$, their L coherently aligned training contributions have exactly depth-independent total coefficient (`sqrtDepthScales_coherentUpdateCoefficient`). The same per-layer contributions have orthonormal-direction quadratic budget proportional to $1/L$ (`sqrtDepthScales_updateQuadraticBudget`); correlation is therefore a load-bearing hypothesis, not a decorative intuition. Holding the effective update scale constant instead makes the coherent squared coefficient grow linearly with depth (`sqrtBranch_constantUpdate_coherent_sq`). Moreover, for nonzero branch amplitude, requiring the coherent coefficient to stay invariant uniquely forces the paired $1/\sqrt{L}$ effective-update scale (`coherent_invariance_forces_sqrt_updateScale`). These exact finite statements do not claim the paper's tensor-program limit, optimizer derivation, infinite-depth feature diversity or empirical transfer.

Fact C.1 and Theorem 4.1 of [65] admit an exact finite interpretation. The activation-gradient pullback through one normalized scalar feature is the gain-to-standard-deviation ratio times the incoming gradient after subtracting its constant-batch component and its component along the normalized activation. The checked Pythagorean identity subtracts the squares of both components explicitly (`exactBatchNorm_gradientPullback_normSq_eq`). The unscaled projection is therefore nonexpansive and is strictly smaller when either removed component is nonzero (`batchGradientProjection_normSq_lt`). This is not universal strict smoothing: a nonzero four-sample gradient orthogonal to both directions is unchanged (`orthogonalFour_projection_unchanged`). The result recovers the finite activation-gradient algebra; it does not imply the paper's Hessian, weight-space, global-optimization or empirical claims.

The same discipline recovers the decision-theoretic core of logit-adjusted softmax. Adding temperature-scaled log prior multiplies unnormalized softmax mass by the corresponding powered prior; at unit temperature, positive class-conditional likelihoods recover the finite Bayes posterior. The flattened finite-window estimator is normalized, and additive smoothing makes every class prior strictly positive. More importantly, the paper's class-balanced decision boundary is proved directly: any pointwise maximizer of class-conditional likelihood minimizes finite class-balanced error (`likelihoodMaximizer_minimizes_classBalancedError`). A normalized two-class fixture with a three-to-one prior separates that predictor from the ordinary sample-risk optimum: their balanced errors are $1/3$ and $1/2$, while their prior-weighted errors are $1/3$ and $1/4$, respectively (`balanced_and_prior_optima_separate`).

Two additional sources constrain the machine above those residual blocks. Scaled dot-product attention is formalized as positive affine weighting, with exact common-score-shift invariance and a multiset counterexample: duplicating one equally scored key changes the readout from 1 to 4/3. The auxiliary routing factor in Switch Transformer is recovered exactly. Its uniform value is one, and Cauchy–Schwarz proves this is minimal when hard dispatch fractions equal mean router probabilities. Under the two-vector definition in the paper, however, an admissible three-expert batch has factor $4653/5000 < 1$; thus the stated uniform-minimum conclusion needs the missing self-consistency premise (`one_le_switchRoutingBalanceFactor_selfConsistent`, `switch_uniform_is_not_global_` minimum). The local-capacity mechanism in GShard has a different, deterministic guarantee: if every independently routed group accepts at most C tokens for an expert, the aggregate accepted load is at most the group count times C. This bound is attained when every group is full. Its converse is false—a two-group load $(2, 0)$ meets the aggregate bound for local capacity one while violating the actual local invariant (totalExpertLoad_le_card_mul_capacity, global_capacity_does_not_imply_local_capacity). The symbolic resource equations of SwitchHead expose a complementary boundary. At fixed head count and projection dimensions, every positive number of active value/output experts has a strictly positive MAC overhead relative to the dense projection model (transformerXLMACs_lt_switchHeadMACs_sameGeometry). Thus expert routing alone is not the source of the advertised saving: the design must reduce the number of attention matrices or alter other dimensions. Under the paper's fused smart-kernel storage model, storage is strictly monotone in head count; paired fixtures show a fourfold declared storage reduction after reducing heads and a MAC increase when all geometry is held fixed (declaredTrainingStorage_lt_iff_heads_lt). These are symbolic cost results, not claims about allocation, wall-clock time or task efficacy. The decaying outer-product memory of Ba et al. is likewise unrolled over arbitrary finite key and value coordinates: its matrix read is exactly chronological decay-weighted dot-product attention. Zero decay keeps only the newest instruction and unit decay recovers scaled additive attention; at decay one-half, the scalar streams $(2, 5)$ and $(5, 2)$ read as 6 and $9/2$, respectively (read_writeAllWithDecay, decayed_write_order_matters). Proposition 1 of [31] is now recovered as an exact finite-dimensional identity. A hand-constructed, unnormalized attention head stores the negative finite-batch gradient change of a linear predictor; negating its query-token target therefore gives exactly the predictor's value after one gradient step (queryTokenTargetAfter_eq_neg_updatedPrediction). At zero initial weight this reduces to the corresponding finite kernel smoother (neg_queryTokenTargetAfter_zeroWeight_eq_kernelSmoother). The restriction to training-token keys and values is substantive: at nonzero initial weight, admitting the zero-target query contributes an additional residual term, as an executable counterexample shows (full_attention_query_contamination). This is a theorem about the paper's explicit linear-attention construction, not evidence that training discovers that construction. The rank-one associative experts of Lu et al. sharpen this memory boundary. A routed rank-one matrix acts exactly as the sum of atomic key–value reads, and a one-hot route recovers one atom (read_routedRankOneMemory). The intrinsic L2 scores have unit squared mass only when their activation energy is strictly positive; zero activation yields zero score mass. Active-set restriction provably bounds nonzero routes by the declared top-k budget. Two executable counterexamples delimit the architecture: duplicate keys make dense routing retrieve both values, and thresholding normalized weights $(3/4, 1/4)$ at $1/2$ leaves mass $3/4$, not one (dense_routing_interferes). Thus atomic memory, sparse selection, and post-threshold renormalization are separate obligations. The continuous modern Hopfield update of Ramsauer et al. gives a complementary retrieval certificate. For any finite score family, a target margin Δ makes each competing softmax weight at most $\exp(-\beta\Delta)$ and bounds total off-target mass by the number of competitors times that factor (offTargetMass_le). Consequently every retrieved coordinate differs from the target by at most its declared pattern diameter times the same mass bound (abs_modernHopfieldRead_sub_target_le). This deterministic theorem generalizes the score algebra without claiming the paper's random-pattern capacity or global fixed-point result. Equal scores retrieve an average rather than either pattern, and zero inverse temperature erases a strict raw-score margin (equal_score_collision_returns_average). Finally, the affine acceleration equations surveyed by Saad are lifted to arbitrary real normed spaces. Sum-to-one weights preserve fixed points and transport error exactly, but signed affine weights can amplify unit-bounded errors by a factor three. This is the reason an accelerated settling profile needs a residual or acceptance safeguard in addition to an affine extrapolator.

The original binary local objective of Hinton's Forward–Forward algorithm [58] now has an exact finite theorem as well. For squared activation goodness $g = \|y\|^2$ and threshold θ, the two local probabilities satisfy

$$p_+(g) = \sigma(g - \theta), \qquad p_-(g) = 1 - p_+(g),$$

and both lie strictly between zero and one (`negativeProbabilityOfGoodness_eq_one_sub`). Their negative-log losses are exactly the corresponding softplus expressions. Differentiating with respect to goodness gives a strictly negative slope for the positive pass and a strictly positive slope for the negative pass (`positiveLoss_derivative_neg`, `negativeLoss_derivative_pos`). Raw squared goodness scales quadratically, whereas normalizing any nonzero activation makes its goodness exactly one (`activationGoodness_l2Normalize_eq_one`). The zero vector remains at goodness zero, and squared goodness alone cannot detect a direction reversal. Finally, a two-negative fixture proves that this original binary objective is not the later full-comparison multiclass objective (`originalBinaryObjective_ne_twoNegativeFullComparison`). These identities recover the local mechanism; they do not establish convergence, task accuracy, biological plausibility, or equivalence to backpropagation.

Changing an error-coordinate frontier has a separate accounting obligation. With positive entry–exit gap δ, total membership churn N, distinct coordinates ever switched D, and summed switch-event variation V, the formal aggregate law is

$$\delta\,(N - D) \le V.$$

The subtraction is exact first-switch accounting, not slack: `totalSwitchEvents_eq_chargeable_add_switchedCoordinateCount` decomposes churn into chargeable switches and one uncharged event per switched coordinate, while `totalChurn_without_distinctCount_cannot_be_charged` refutes charging every event. A preregistered single-checkpoint diagnostic therefore remains a negative rather than a training license: fixed 5%/10% hysteresis made 2,761 changes and captured less final squared mass than pointwise top-5%, but did not log D or per-coordinate variation. The next trace schema must expose (N, D, V) before the aggregate theorem can certify a dynamic frontier.

An audit need not create a duplicate module. The fixed-point residual and implicit gradient in Theorem 1 of Deep Equilibrium Models match the existing affine workspace bridge: under spectral radius below one, the local equilibrium branch has derivative $(I - L)^{-1}$ composed with the parameter drive (`linearDEQImplicitEquilibrium_hasStrictFDerivAt`). The atlas records this as an affine complex-normed-space specialization, not as a proof of the paper's nonlinear theorem or of a constant-memory runtime.

The same distinction matters for higher-order credit. One inner update from model-agnostic meta-learning is formalized for arbitrary differentiable scalar training gradients and evaluation losses. The exact outer derivative contains the adaptation-Jacobian factor $1 - \alpha H$. Zero rate and zero curvature recover the first-order approximation, whereas a quadratic fixture with $\alpha = 1$ and $H = 2$ makes the exact derivative 1 and the first-order approximation -1 (`oneStepMetaObjective_hasDerivAt`, `firstOrder_metaGradient_reverses_direction`). Thus first-order meta-learning is an explicitly delimited approximation, not an interchangeable implementation of the exact update.

4 The belief axis: fusion is addition

Write belief in *natural coordinates*: evidence counts (n^+, n^-) in the discrete case, the information form $(\eta, \Lambda) = (\Sigma^{-1}\mu, \Sigma^{-1})$ in the Gaussian case. Then Bayesian fusion of independent evidence is *addition*, and the familiar update rules are that addition seen in another chart.

Theorem 4.1 (Fusion is addition; charts are shadows). *Gaussian fusion, probabilistic-logic revision in weight-of-evidence form, and revision on binary counts are the same operation (`gaussianFusion_eq_plnRevisionWTV`, `gaussianFusion_eq_binaryCounts_revision_strength`). The settled state of a linear-Gaussian relaxation is the posterior mean (`equilibrium_iff_eq_posteriorMean`), so "settling performs inference" is a theorem rather than a slogan.*

The practical reading is that the optimal gate in a gated-interpolation cell is the Kalman gain: a learned mixer can at best match hardwired addition when the measurement model is calibrated, never beat it. Learning is therefore necessary in the *measurement* maps and the *gain schedule*, not in the fusion step.

That optimality is conditional, and each condition fails somewhere real. Addition never forgets, so under drift it is provably suboptimal and the correct old-information retention is the derived rate $P/(P+Q)$ rather than a tuned constant. Naive addition double-counts correlated re-observation, and the overstatement is exactly the declared overlap. A single natural-coordinate slot projects away multimodal posteriors. And a miscalibrated measurement model poisons exact fusion, though recalibrating the measurement map reproduces the optimal update with fusion left hardwired—so every fusion-stage repair relocates to the measurement map. These four boundaries are the entire content of "when should the belief axis stop being additive".

5 The state axis: typed slots, routing, and what stays safe

Moving from one recurrent vector to addressed slots adds structure and gives away freedom. The licensing condition is sharp: independent per-slot treatment is correct exactly when the cross-slot coupling vanishes, and coupled slots require a joint correction rather than a per-slot approximation.

A command-routed workspace—opaque control tokens selecting a soft mixture of reusable read–transform–gate–write operators—sits on the same axis. Three results delimit it.

Theorem 5.1 (Routing buys selectivity, not a new update class). *A simplex mixture of gated-interpolation experts is again a single gated interpolation under the mixed gain (`mixedExpertStep_eq_singleInterpolation`). Command tokens carry no semantics of their own: trajectories are determined by the induced routing schedule alone (`commandTrajectory_eq_of_mixtureSchedule_eq`). Immutable per-slot evidence survives every routing schedule, temperature and depth (`runTripleSlotSchedule_evidence`).*

Theorem 5.2 (Compositionality is commutation). *Swapping two affine phases shifts the state by $[B,A]x+(Ba+b-Ab-a)$ (`affineCompositionDiscrepancy_exact`); order-independence holds if and only if the linear parts commute* and *the affine obstruction vanishes (`affinePhases_orderIndependent_iff`). Commuting linear parts alone do not suffice—a one-dimensional bias counterexample closes that loophole. Equivalently the pairwise interference energy $\|[A,B]\|_F^2$ must vanish (`linearPhases_orderIndependent_iff_interferenceEnergy_zero`).*

Theorem 5.3 (Adjoint pairing gives exact time symmetry). *For a group-valued reversible step family, taking the adjoint reverses both the step parameter and the execution order. The half-step composition with its adjoint is time-symmetric (`adjointPair_timeSymmetric`), and every palindromic scaled composition of a time-symmetric stage remains time-symmetric without assuming that its stages commute (`palindromic_scaledStageComposition_timeSymmetric`). This is the exact algebraic content of the adjoint and symmetric-composition construction of Blanes, Casas, and Murua [57]. A three-state permutation fixture shows that deleting the closing stage of a noncommuting palindrome can destroy symmetry (`nonpalindromic_noncommutingComposition_not_timeSymmetric`); conversely, commuting translations show that a syntactic palindrome is sufficient but not necessary (`nonpalindromic_can_still_be_`* *timeSymmetric). These group laws do not by themselves establish numerical order, convergence, symplecticity, or invertibility of a learned phase.*

Theorem 5.4 (Stability does not compose)**.** *Two commands that each individually extinguish in two steps can alternate to 4^n growth (individuallyStable_switchingDiverges). Per-command guarantees are therefore insufficient for command* programs*; the honest certificate is schedule-uniform, and commuting stable generators always admit a common quadratic Lyapunov function (commutingCommonQuadraticLyapunov_crown).*

Theorem 5.5 (Finite-word growth is the executable joint-spectral layer)**.** *The nonlinear joint spectral radius of Deidda, Guglielmi, and Tudisco [33] takes worst-case growth over every switching word at each depth. At the finite layer, a uniform energy factor for all words of length m and one for all words of length n compose multiplicatively at length $m + n$ (FiniteWordEnergyBound.add). Hence any strict block factor gives a geometric envelope for every arbitrary schedule whose length is a multiple of the block depth (FiniteWordEnergyBound.geometricEnvelope_tendsto_zero). This is more expressive than demanding contraction at each step: one nilpotent command has no strict one-step factor but has a zero two-step factor (expandingThenExtinguishing_no_strictOneStepBound, expandingThenExtinguishing_twoStepBound). Conversely, checking each command only under self-repetition is unsound for switching, as the alternating counterexample has no strict two-step uniform factor (divergentBoolTransition_no_strictTwoStepBound). The formal result is the finite, checkable substrate; it does not claim the paper's cone-norm limsup construction, Thompson-metric estimate, or asymptotic-stability equivalence.*

Theorem 5.6 (Two populations compile a directed graph exactly)**.** *The construction of Ashwin and Postlethwaite [34] associates one state cell to every vertex and one transition cell to every edge. At the finite wiring layer, a state-to-transition coupling followed by a transition-to-state coupling realizes exactly the source graph relation (twoPopulationTransition_iff_graphTransition); a three-state cycle is checked as a positive fixture (threeCycle_twoPopulation). Parallel edges remain distinct transition cells.*

The address map is load-bearing. Merging transition cells cannot delete a source transition, and an injective address map preserves exact recovery (compressedPopulationTransition_iff_graphTransition_of_injective). But a concrete noninjective map cross-wires the source half of one edge with the target half of another, producing an absent transition (constantTransitionAddress_creates_spurious_cross). Thus a routed carrier must retain edge identity or declare and audit its compression semantics. This theorem deliberately stops at discrete incidence: it does not prove the paper's ODE-level heteroclinic or excitable realization, robustness, attraction, dwell time, noise behavior, parameter-region openness, or excitation thresholds.

Theorem 5.7 (GLV rates certify local itinerary direction and passage)**.** *The generalized Lotka–Volterra construction of Voit and Meyer-Ortmanns [35] gives a rare invader j at a single-species resident equilibrium i the local rate*

$$\nu(A_{ji}) = \rho - A_{ji}\rho/\gamma.$$

The full finite-species GLV field has the single-species state as an exact equilibrium, and its invader coordinate on a resident–invader probe reduces exactly to the scalar restriction (glv_singleSpeciesState_is_equilibrium, glvVectorField_residentInvaderState_invader). That restriction has derivative $\nu(A_{ji})$ at zero (glvInvaderField_hasDerivAt_zero). For positive reproduction and death rates, expansion is equivalent to $A_{ji} < \gamma$, contraction is equivalent to $A_{ji} > \gamma$, and a smaller predation rate is exactly a larger invasion rate (glvInvasionRate_pos_iff, glvInvasionRate_neg_iff, glvInvasionRate_order_iff). The source's stronger incoming condition is also recovered in both directions:

$$\nu(e) < -\nu(c) \iff 2\gamma - e < c$$

(`glvContractionDominates_iff`).

Consequently the source inequalities package into one named certificate: both outgoing directions expand, the small-cycle direction is faster, both incoming contractions dominate it in magnitude, and all other transverse directions contract (`GLVLocalDirectionProfile.certified_directionality`). The representative decimal parameters are evaluated as exact rationals, yielding $87/107$, $77/107$, $-93/107$, *and* $-18/107$ *for the four rate classes (`standardHierarchicalGLVProfile_exact_rates`). More importantly, this analytic layer meets the finite compiler: away from the radial coordinate, a three-cycle graph edge exists if and only if its assigned local invasion rate is positive (`threeCycle_transition_iff_positive_invasion`). Assigning a transverse predation rate below death supplies the executable opposite boundary (`below_death_transverse_direction_is_unstable`).*

For the linearized abundance $u(t) = u_0 e^{\lambda t}$, *the threshold time* $\log(\theta/u_0)/\lambda$ *reaches* θ *exactly and decreases strictly with the positive initial amplitude (`linearizedAbundance_at_exitTime`, `linearExitTime_strictAnti_initial`). A nonpositive local rate cannot reach a higher threshold at nonnegative time (`nonpositive_rate_cannot_reach_higher_threshold`). These are local direction and passage laws, not a proof of a global heteroclinic orbit, network attraction, stochastic dwell-time scaling, perturbation robustness, or realization by the trained routed carrier.*

Theorem 5.8 (Local graph refinement preserves command coverage)**.** *The four-edge forward lift of Ninite and Jungers—splitting one phase into one copy for each outgoing neighbour—realizes every nonempty finite command word whenever the original graph does (`forwardLift_graphPathComplete`). Every phase-indexed regional Lyapunov certificate pulls back through the lift projection with exactly the same contraction rate (`PathCompleteRegionalLyapunov.forwardLift_rate`). This preserves certificate feasibility; it does not assert that a chosen lift strictly improves an optimized bound. Conversely, lifting a graph that lacks one command does not manufacture that command (`falseOnlyUnitGraph_forwardLift_not_pathComplete`).*

Against all of this, safety is indifferent. For arbitrary routes, temperatures, schedules and depths, ranked acceptance coincides with checker acceptance and recall of accepted outputs is unchanged (`inheritanceCrown`). When legality is owned by an external checker and the network only orders legal choices, the entire state axis lives on the safe side by construction, and the theorems above are about competence rather than safety.

For readers arriving from the transformer literature, the whole state axis is a familiar machine with one substitution. Attention is the *read* at every point—the encoder, the state carriers, the action scorer—and what varies across the axis is only the *write*:

mechanism	transformer	CAROM	this work
read	multi-head attention	content-addressed read	same, at three sites
transform	feed-forward block	per-operator ϕ_k	per-operator maps
write	residual + norm	gated residual	gate / residual / *addition*
state	residual stream, KV cache	fixed-address slots	typed slots or (n^+, n^-)
routing	mixture-of-experts gate	command-conditioned mix	selectivity, same class
recurrence	weight-tied loop	T recurrent updates	settling, interior-optimum depth
no-weight adaptation	in-context learning	workspace contents	evidence registers, exact

The last column's entries are not analogies but theorems: routing stays in the gated-interpolation class, the belief write is exactly Bayesian fusion in natural coordinates, the recurrence depth has a proved interior optimum, and the evidence registers are an exact instance of the fast-state model of §11—the one member of the test-time-adaptation family whose calibration is provable rather than empirical.

6 The unified carrier: one machine, not four rivals

The axis decomposition is not an instruction to keep four separate decoders forever. Ask what each component actually is, in the vocabulary of a machine. The recurrent vector is a control register: cheap, sequential, decisive. The typed workspace is addressed memory: contents at fixed, elaborator-owned locations. The belief registers are the memory's *metadata*: how much evidence stands behind each cell, and therefore how fast it should be overwritten — the role error-correction bits and replacement policies play in hardware. The routed operators are a small shared arithmetic unit. A hardware designer would never run a tournament among these; the interesting object is the machine that has all of them, each doing its own job. The unified carrier places them in one product state (Figure 1).

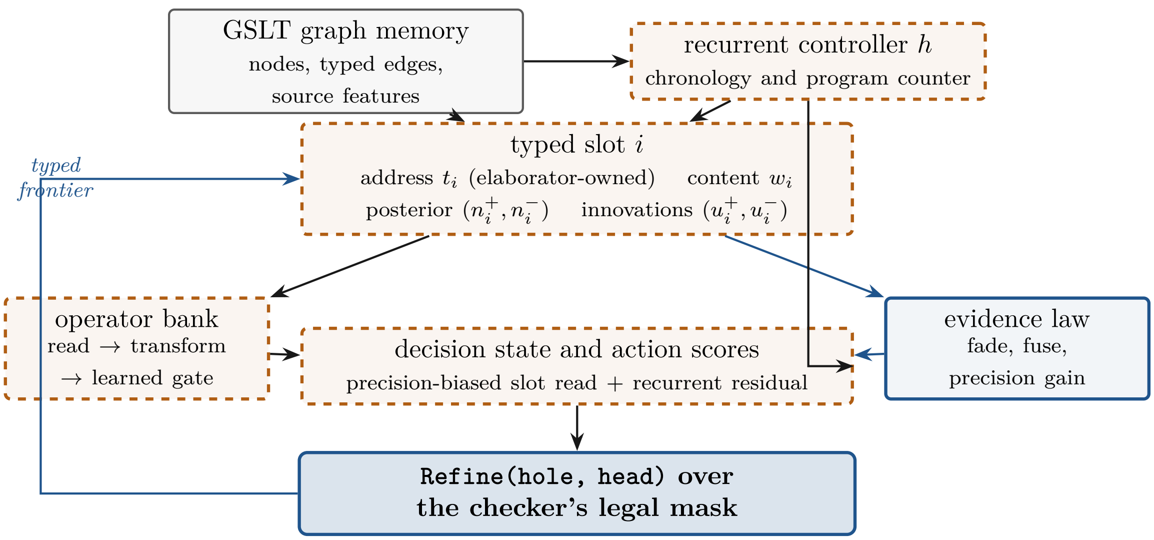

Figure 1: The unified carrier as one machine. Solid blue borders mark components whose laws are machine-checked (the evidence law and the typed-action waist); dashed orange borders mark learned components. The address plane stays elaborator-owned: no learned state can enlarge the legal action support.

For each typed slot i, let

$$(w_i, n_i^+, n_i^-, u_i^+, u_i^-, t_i)$$

denote content, posterior evidence, episode-local innovations and an elaborator-owned type signature. A compact recurrent state h carries chronology. Writing the featurewise precision vector $\Lambda_i = n_i^+ + n_i^-$ and denoting its feature mean by an overbar, the derived retention vector and scalar content gain are

$$r_i = \frac{1}{1 + \sigma^2_{\text{jump}} \Lambda_i}, \qquad \gamma_i = \frac{\bar{\Lambda}_i^{\text{fresh}}}{\overline{r_i \odot \Lambda_i} + \bar{\Lambda}_i^{\text{fresh}}}.$$

Operator k proposes a candidate c_{ki}, learned gate g_{ki} and nonnegative fresh evidence packet. One settle step is

$$n_i^\pm \leftarrow r_i n_i^\pm + \text{fresh}_i^\pm, \qquad u_i^\pm \leftarrow u_i^\pm + \text{fresh}_i^\pm,$$
$$w_i \leftarrow w_i + \frac{1}{K} \sum_k \gamma_i g_{ki} (c_{ki} - w_i), \qquad h \leftarrow \text{GRU}(x_{\text{carrier}}, h).$$

Thus the learned workspace gate chooses *what* to write while the precision ratio prices how much the current evidence licenses it. When old precision is zero and fresh precision is positive, $\gamma_i = 1$ and the content update is exactly the workspace write (`contentStep_zeroOld_eq_workspace`). When both are zero, the implementation-defined gain is zero; this boundary prevents a division-by-zero "workspace collapse" from being asserted without evidence.

Precision-routed reads. For operator k, slot j receives

$$\ell_{kj} = \frac{\langle q_k, k_j \rangle}{\sqrt{d}} + \beta_k \log(1 + \bar{\Lambda}_j), \qquad \beta_k = b_k^2 \geq 0.$$

The exact identity

$$e^{\ell_{kj}} = e^{\langle q_k, k_j \rangle / \sqrt{d}} (1 + \bar{\Lambda}_j)^{\beta_k}$$

is proved in `exp_precisionBiasedLogit`; positive β_k makes pairwise odds strictly increase with precision, while $\beta_k = 0$ recovers the workspace read. This is an algebraic reliability bias. It is not called a posterior-predictive distribution without separately declaring a probabilistic model.

Persistent evidence without double counting. A fresh hole may initialize its evidence from a persistent table keyed by the injective tuple (role, parent operator, argument). At an episode boundary the table decays once and adds only the innovations $u^{\pm}$ produced during that episode. Reabsorbing the posterior $n^{\pm}$ would import the prior a second time; the development proves that error on a concrete fixture (`posterior_reabsorption_doubleCounts`). Batch absorption is intentionally outside the causal interface because episode order changes the decay. The persistent feature is disabled by default until an experiment declares how episodes are ordered.

Continuous opinions retain uncertainty separately from prediction. The K-class evidential chart of Sensoy, Kaplan, and Kandemir [70] starts from nonnegative evidence e_i, sets $\alpha_i = e_i + 1$ and $S = \sum_i \alpha_i$, and assigns

$$b_i = \frac{e_i}{S}, \qquad u = \frac{K}{S}, \qquad \mathbb{E}[p_i] = \frac{\alpha_i}{S}.$$

The finite formalization proves $u + \sum_i b_i = 1$, that the posterior means sum to one, and the exact decomposition $\mathbb{E}[p_i] = b_i + u/K$ (`uncertainty_add_sum_belief`, `posteriorMean_eq_belief_add_uniform_uncertainty`). Fusion must add raw evidence: it increases strength only by the fresh evidence mass. Adding two posterior concentration vectors instead imports the unit base concentration twice, overcounting exactly one unit per class (`strength_fuse_eq_strength_add_totalEvidence`, `sum_concentration_add_eq_strength_fuse_add_classCount`). Positive fresh evidence strictly lowers u. Conversely, uniformly rescaling all concentrations preserves every posterior mean while dividing uncertainty by the scale (`posteriorMean_rescaleConcentration`, `uncertainty_rescaleConcentration`). A checked binary fixture makes the boundary concrete: zero evidence and balanced high conflict both predict $(1/2, 1/2)$, but have uncertainties 1 and $1/10$ (`binary_sameMean_differentUncertainty`). Thus the evidence metadata plane must carry strength rather than reconstruct it from logits. These identities do not certify calibration, out-of-distribution detection, adversarial robustness, causal packet identity, or task efficacy.

Predictive means do not determine the kind of uncertainty. The Prior Network decomposition of Malinin and Gales [71] is recovered for a finite weighted ensemble of categorical predictors. If $\bar{p} = \sum_j w_j p_j$, then

$$I = H(\bar{p}) - \sum_j w_j H(p_j) = \sum_j w_j D_{\mathrm{KL}}(p_j \| \bar{p}) \geq 0, \qquad I \leq H(\bar{p}).$$

The formalization connects the categorical divergence explicitly to the existing Knuth–Skilling KL divergence (`categoricalDivergence_eq_ksKLDivergence`). The average-divergence identity is `distributionalInformation_eq_averageDivergenceFromMean`. Its nonnegativity and total-entropy bound are `distributionalInformation_nonneg` and `distributionalInformation_le_totalUncertainty`. The zero boundary for duplicate predictors is (`duplicateEnsemble_distributionalInformation_eq_zero`). A binary negative fixture makes the design consequence exact. Separated point masses and two duplicate uniform predictors both have mean $(1/2, 1/2)$; the first assigns $\log 2$ to distributional information and zero to expected data uncertainty, while the second makes the opposite assignment (`sameMean_oppositeUncertaintyDecomposition`). Consequently logits or posterior means alone cannot determine evidence routing: the metadata plane must retain either the predictive distribution over predictors or an independently justified sufficient statistic. This finite result does not derive the paper's Dirichlet digamma expectation or differential-entropy formula, and it does not certify calibration, out-of-distribution detection, accuracy, or task efficacy.

Expert expansion has two distinct normalization steps. For task-free expert growth, Lee et al.'s neural Dirichlet-process mixture [72] assigns existing score $s_k = N_k \ell_k$ and candidate score $s_0 = \alpha \ell_0$, then normalizes over all $K+1$ choices. The finite formalization proves unit total responsibility (`sum_existingResponsibility_add_newResponsibility`), that the candidate is maximal exactly when its raw score is maximal (`candidateMaximal_iff_rawScoreMaximal`), and that adding the candidate preserves every pairwise existing-expert odds ratio (`existingResponsibility_preserves_pairwiseOdds`). If the candidate is rejected, Algorithm 1 performs a second normalization over existing experts. That step cancels the candidate score exactly (`renormalize_combinedResponsibility_eq_conditional`) and makes the stored-count update add one observation (`conditionalExisting_update_adds_one`). It cannot be omitted: with existing scores $(2, 1)$ and candidate score 1, the combined responsibilities are $(1/2, 1/4, 1/4)$, so updating only existing counts without conditioning adds $3/4$ rather than one (`twoExpert_missingRenormalization_loses_mass`). This gives a precise exactly-once obligation for any future dynamic operator pool. It does not establish calibrated neural likelihoods, true task recovery, validity of the sequential variational approximation, short-term-memory homogeneity, bounded expert growth, or efficacy.

Causal transport across frames. The innovation ledger says which evidence may be counted; it does not by itself say whether two coordinate frames express the same mechanism. Following the metric-aware transport contract of Goertzel's Causal Memory and Credit Protocol, let $T : X \to Y$ connect source operators H_a^X to memory operators H_a^Y. The development separates the two required conditions:

$$H_a^Y T = T H_a^X, \qquad \langle Tx, Ty \rangle_Y = \langle x, y \rangle_X.$$

Exact intertwining extends to every finite operator word (`intertwines_operatorWord`); approximate local equations extend with a recursive budget that charges target-side amplification of earlier error and source-side growth of later error (`approxIntertwines_operatorWord`). Under metric preservation, every transported word has the same pairwise response (`operatorWord_inner_preserved`). The separation is essential: the zero map intertwines *every* pair, including two unequal scalar operators, but fails metric preservation (`zeroTransport_intertwines_unequal_scalarOperators`). Finally, any additive evidence-frame map commutes with ledger assimilation (`mapEvidence_assimilate`), so a fresh causal identity contributes one transported payload rather than one payload per manifestation. This theorem does not infer the transport or causal identity from data; those remain measured inputs.

Direction is not evidence. The same protocol distinguishes a new mechanism direction from new evidence about an existing mechanism. Let F_{cc} be a current-score covariance, F_{MM} a positive-definite stored-score covariance, and F_{cM} their cross covariance. The remaining conditional information is

$$F_{c|M} = F_{cc} - F_{cM} F_{MM}^{-1} F_{Mc}.$$

Positive semidefiniteness of the joint block covariance is equivalent to positive semidefiniteness of this Schur complement (`jointCovariance_posSemidef_iff_conditionalInformation`), and the removed predictable component is itself positive semidefinite. Hence $0 \preceq F_{c|M} \preceq F_{cc}$ (`conditionalInformation_between_zero_and_current`). An exact linear duplicate has zero residual and zero conditional information, whereas a zero-cross-covariance observation may occupy an already represented direction and retain its full information (`independent_sameDirection_fullInformation`). Half correlation retains three quarters in the scalar unit fixture. The source uses a Moore–Penrose inverse for singular stored covariance; the proved theorem deliberately covers the positive-definite specialization, and an explicit singular fixture prevents Lean's totalized matrix inverse from being mistaken for a pseudoinverse certificate.

Refinement is exact and nontrivial. Under an explicit mode, parameter relation and state projection, the unified carrier simulates the typed workspace for every finite policy trajectory (`workspaceEndpoint_policyTrajectories_eq`). This is not validation by definition: positive old precision separates the full carrier from that endpoint (`positivePrecision_fullCarrier_separates_workspace`), and two equal workspace/evidence snapshots with different recurrent histories can produce different policies (`recurrentHistory_fullPolicy_separates`). The common masked source policy remains supported exactly at legal operators, with an illegal-operator fixture (`illegalOperator_stays_absent_after_unifiedStep`). The carrier may reorder legal actions; it cannot widen the language.

When does the metadata plane earn its cost? In the scalar linear-Gaussian jump model, the exact one-revision risk reduction is $R^2/(P+Q+R)$ and decreases strictly with jump variance. Including a one-time activation cost and recurring metadata cost yields a decidable revision-count threshold (`retentionNetAdvantage_pos_iff_revisionThreshold`). The same file proves a low-drift positive fixture and a high-drift negative fixture. This says what the architecture experiment must measure—revision count, drift and work in common cost units—rather than predicting that persistent belief will win.

Fast/slow consolidation has an exact conditional rate. The slow learner of DualNet [37] uses the Lookahead [36] update: starting from a slow state x, take K fast steps with map S and then set

$$L_{\beta,K}(x) = x + \beta(S^K(x) - x).$$

Every fixed point of S is a fixed point of $L_{\beta,K}$ (`slowUpdate_fixed`). If the fast step obeys the explicit pointwise certificate $\|S(x) - x_\star\| \le q\|x - x_\star\|$, then

$$\|L_{\beta,K}(x) - x_\star\| \le \left(1 - \beta + \beta q^K\right)\|x - x_\star\|$$

for $0 \le \beta \le 1$ (`fastIterate_error_le`, `slowUpdate_error_le`). The factor is strictly below one for a positive interpolation, a genuinely contractive q, and $K > 0$; when $0 < q < 1$, every additional fast step strictly lowers the one-cycle factor (`lookaheadRate_lt_one`, `lookaheadRate_strictly_decreases`). For an affine contraction, the same factor gives the exact trajectory across arbitrary slow cycles (`repeated_slowUpdate_affine`).

Both opposite boundaries are executable. Setting $\beta = 0$ discards all fast progress, while $\beta = 3$ turns the perfect fast map $x \mapsto 0$ into a factor-two expansion (`zero_interpolation_discards_fast_progress`, overrelaxation_can_expand). Thus this profile may be promoted only after traces bind K, β, a reference state, the observed contraction premise and elapsed work. The theorem does not infer contraction for stochastic optimization or predictive-coding settling, and decreasing the one-cycle factor does not by itself make a larger K work-normalized optimal.

A fixed-representation control has exact finite geometry. The RanDumb baseline of Prabhu et al. [38] freezes a normalized cosine–sine random Fourier map and updates class statistics online. For phase vectors $a, b \in \mathbb{R}^D$, its finite feature inner product reduces exactly to

$$k_D(a,b) = \frac{1}{D}\sum_{i=1}^{D}(\cos a_i \cos b_i + \sin a_i \sin b_i) = \frac{1}{D}\sum_{i=1}^{D}\cos(a_i - b_i)$$

(rffSimilarity_eq_average_cos_sub). For every nonempty feature bank, $k_D(a,a) = 1$, k_D is symmetric, and $-1 \le k_D(a,b) \le 1$ (rffSimilarity_self, rffSimilarity_comm, rffSimilarity_mem_Icc). Normalization is not injectivity: one-dimensional phases 0 and 2π are distinct but have similarity one (zeroPhase_ne_twoPiPhase, periodic_phase_collision).

The count-weighted class-mean update has the exact conservation law

$$(n+1)\mu_{n+1} = n\mu_n + x_{n+1},$$

and its recursive form counts every finite sample exactly once (onlineMeanUpdate_balance, length_smul_onlineMean_eq_sum). This makes a frozen random representation a characterized future control for the carrier experiment, not evidence that it will win. The theorem does not establish probabilistic RBF approximation, covariance invertibility or shrinkage quality, Mahalanobis-classifier accuracy, or the source's empirical comparison with learned representations.

Streaming covariance has a certified precision domain. The Deep Streaming LDA update of Hayes and Kanan [39] maintains a shared covariance by averaging its previous value with a count-scaled rank-one innovation. The rank-one contribution is positive semidefinite, and the complete streaming update preserves positive semidefiniteness (scatterIncrement_posSemidef, covarianceUpdate_posSemidef). For any incoming $\Sigma \succeq 0$, the source's shrinkage operator

$$\widetilde{\Sigma}_\varepsilon = (1-\varepsilon)\Sigma + \varepsilon I$$

is positive definite, and hence invertible, throughout the explicit domain $0 < \varepsilon \le 1$ (shrinkCovariance_posDef, shrinkCovariance_isUnit). Both endpoints matter: zero shrinkage does not repair a singular zero covariance, while extending the coefficient beyond one can assign a negative coefficient to the incoming covariance and destroy positive semidefiniteness (zero_shrinkage_zeroCovariance_not_isUnit, excessive_shrinkage_can_destroy_posSemidef). These results certify the online covariance geometry and the admissible precision domain; they do not establish representation quality, statistical consistency, calibration, classification accuracy, or the paper's empirical comparisons.

Covariance-aware prototypes need their Gaussian normalizer. FeCAM [40] classifies frozen features by class prototypes and covariance-aware distances. For a positive-semidefinite precision matrix Λ, the formalized Mahalanobis quadratic

$$D_\Lambda(x,\mu) = (x-\mu)^\top \Lambda (x-\mu)$$

is nonnegative, and $\Lambda = I$ recovers squared Euclidean distance (mahalanobisSquared_nonnegative, mahalanobisSquared_identity). With equal priors and one shared positive scalar precision, maximizing the corresponding Gaussian density is exactly equivalent to minimizing D_Λ (`same_precision_density_gt_iff_mahalanobis_lt`).

Class-dependent covariance changes that conclusion: the Gaussian density also contains the determinant-dependent normalization. A scalar fixture with precisions 1 and 9 makes a Mahalanobis-only rule choose the first class while the normalized Gaussian score chooses the second (`heterogeneous_precisions_reverse_distance_decision`). Nor does correlation normalization remove the issue: two positive-definite 2×2 correlation matrices can have identical unit diagonals and different determinants (`correlation_normalization_does_not_equalize_gaussian_normalizer`). Thus the shared-covariance Bayes reduction is exact under its stated normalization hypotheses, whereas a per-class Mahalanobis-only score is not the full Gaussian Bayes classifier in general. These theorems make no claim about feature Gaussianity, covariance-estimation consistency, calibration, classification accuracy, or the source's empirical comparisons.

Frozen projected sufficient statistics are order-invariant. RanPAC [42] freezes a nonlinear random projection and accumulates the projected Gram matrix $G = \sum_i h_i h_i^\top$ together with the class-prototype matrix $C = \sum_i h_i y_i^\top$. Both ledgers compose exactly across concatenated streams and are invariant under every permutation of the same projected observations (`gramLedger_append`, `prototypeLedger_append`, `ledgers_eq_of_perm`). Every finite Gram ledger is positive semidefinite; consequently $G + \lambda I$ is positive definite and invertible for every $\lambda > 0$, without a full-rank premise on the samples (`gramLedger_posSemidef`, `ridgeGram_posDef`, `ridgeGram_isUnit`).

The least-squares link is also exact in the scalar specialization. Completing the square proves that the source's closed form uniquely minimizes the regularized squared-error objective whenever the quadratic coefficient is positive (`scalarRidgeLoss_sub_solution_eq`, `scalarRidgeSolution_minimizes`, `scalarRidgeSolution_unique`). The hypotheses cannot be dropped: using different projectors at two stream positions makes the Gram result depend on raw arrival order, and a negative ridge that cancels the Gram coefficient makes totalized division return a value that is not a minimizer (`changing_projector_breaks_raw_order_invariance`, `singular_negative_ridge_totalizedSolution_not_minimizer`). Thus the theorem licenses a frozen-statistic control only after projector and packet identity are bound; it does not establish random-feature utility, separability, generalization, classifier accuracy, or the source's empirical comparisons.

Analytic incremental ridge is exactly joint under fixed geometry. ACIL [43] maintains the inverse $R = A^{-1}$ of an accumulated ridge normal matrix and, for a new frozen-feature batch (X, Y), forms the batch Schur inverse

$$S = (I + XRX^\top)^{-1}, \qquad R^+ = R - RX^\top SXR.$$

With explicit two-sided inverse contracts for R and S, the matrix development proves that R^+ is a two-sided inverse of $A + X^\top X$ (`updatedInverse_twoSided`). If the old cross statistic is C, the source's residual correction

$$W^+ = W + R^+ X^\top (Y - XW), \qquad W = RC,$$

is exactly the joint ridge head $R^+(C + X^\top Y)$ and satisfies its enlarged normal equation (`recursiveHeadUpdate_eq_joint`, `updatedNormal_mul_recursiveHeadUpdate`). The proof is finite-dimensional and does not require the target columns of successive batches to denote disjoint classes. F-OAL [44] uses the same Woodbury and residual recursions online. It is therefore an instance of this theorem after all labels are embedded in one fixed output coordinate space; padding a stored head with zero columns is not identified with changing the output index type.

Both normal and cross statistics commute across batch order in one fixed feature coordinate system (`updatedNormal_order_independent`, `updatedCross_order_independent`), but the premises are substantive. Exact scalar fixtures show that changing the ridge between phases changes the comparator and that applying position-indexed feature maps can destroy raw-order invariance (`changing_ridge_breaks_joint_equivalence`, `changing_feature_map_breaks_raw_order_invariance`). A singular zero normal matrix fails the inverse contract. Moreover, a singleton inverse-normal statistic determines the exact squared feature magnitude (`featureSquare_recoverable_from_inverseNormal`). Consequently the result licenses an exact frozen-head control and an incremental sufficient-statistic implementation; it does not prove end-to-end equivalence, generalization, accuracy, or privacy merely from discarding exemplars.

Low-rank truncation changes interpolation into projection. LoRanPAC [46] first displays a minimum-norm problem subject to exact interpolation after truncating the feature SVD, but later identifies its classifier as a minimizer of unconstrained squared residual. The distinction is substantive. Let P be the orthogonal projector onto the preserved right-singular subspace, let y be a target row, and let $z = Pz$ be any representable prediction. The formal development proves

$$\|z - y\|^2 = \|z - Py\|^2 + \|Py - y\|^2.$$

Thus Py uniquely minimizes squared loss among representable predictions, whereas exact interpolation is feasible iff $Py = y$ (`squaredLoss_decomposition`, `squaredLoss_eq_zero_iff`, `projectedPrediction_unique_minimizer`, `exists_exact_representable_iff`). A proper rank-one projector preserves a nonzero first-axis target while a nonzero discarded-axis target admits no exact representable prediction (`keepFirstProjector_proper`, `discardedTarget_no_exact_representable_prediction`). This repairs the formulation boundary without refuting the least-squares algorithm; the continual approximate-SVD recurrence and its spectral, generalization and runtime bounds remain separate obligations.

Streaming subspace sketches have two error ledgers. Proposition 1 of Continuous Subspace Optimization [47] first truncates every projected-gradient packet and then applies Frequent Directions to their stream. Its final covariance-sketch bound therefore adds two quantities: the packetwise tail-singular-value energies and the residual of the sketch relative to the aggregate truncated stream. Lean generalizes this composition step from matrices to every normed additive group:

$$\left\|\sum_t A_t - S\right\| \le \sum_t \|A_t - \widetilde{A}_t\| + \left\|\sum_t \widetilde{A}_t - S\right\|$$

(`twoStageSketch_error_le`, `tailEnergy_add_sketchResidual_bounds_final`). A scalar fixture attains both triangle bounds exactly. Conversely, opposing packet errors can cancel, so their summed budgets need not equal aggregate error, and perfect packetwise approximation does not control an uncertified downstream sketch (`opposing_packet_errors_make_local_sum_strict`, `zero_local_error_does_not_bound_bad_sketch`). The SVD tail identity and the Frequent-Directions theorem remain explicit certificate inputs; this result neither re-proves their recurrences nor licenses the paper's accuracy, memory, or runtime claims.

Invariant risk has a fixed-scalar first-order test. Theorem 4 of invariant risk minimization [48] characterizes linear simultaneous optima by environmentwise gradient orthogonality. The formal version works in an arbitrary real Hilbert predictor space. If a classifier w minimizes every environment risk over the image of a continuous linear representation Φ, then

$$\langle \Phi w, \nabla R_e(\Phi w)\rangle = 0 \qquad \text{for every environment } e$$

(simultaneousMinimizer_implies_predictor_gradient_orthogonal). Conversely, if a predictor v satisfies $\langle v, \nabla R_e(v)\rangle = 0$ for every e, the rank-one representation $\Phi_v(a) = av$ makes the fixed scalar classifier $a = 1$ globally optimal for every risk. Hence the fixed-scalar condition is an exact iff, not merely a stationary-point heuristic (scalarOne_simultaneouslyMinimizes_iff_gradient_orthogonal).

The convexity input is explicit: each risk carries its exact gradient and global first-order supporting inequality. A nonzero first-axis predictor passes two opposed transverse affine risks, while an aligned nonzero affine gradient makes scalar one fail global optimality (transverse_affineRisks_scalarOne_simultaneouslyMinimizes, aligned_affineRisk_scalarOne_not_minimizer). This result does not establish causal identification, out-of-distribution transfer, finite-penalty IRMv1 optimization, or the source's later general-position theorem.

Finite environment counts leave a geometric blind subspace. Rosenfeld, Ravikumar, and Risteski [49] show that invariant-risk identification has a nuisance-dimension threshold. The formal tranche isolates two exact linear-algebraic mechanisms from their proof. If $E < d_e$, every linear map from d_e nuisance coordinates to E scalar environment constraints has a nonzero kernel (exists_nonzero_environment_invisible_direction). This is a strict inequality: at $E = d_e$, the identity constraint has no nonzero invisible direction (equal_environment_identity_has_no_nonzero_invisible_direction).

The normalized minimum-norm part of their Lemma C.1 is independent of coordinates. Let $q \neq 0$ be the minimum-norm solution at scalar right-hand side one, and decompose a unit solution as $sq + z$, where $s \geq 0$ and $z \perp q$. Then

$$(s\|q\|)^2 + \|z\|^2 = 1, \qquad s \leq \frac{1}{\|q\|}.$$

Equality forces $z = 0$, so the maximal normalized vector is unique (orthogonalUnitSolution_scale_le, orthogonalUnitSolution_vector_eq_canonical_of_scale_eq). The constraint-level certificate needs only that q maps to the target and is orthogonal to the homogeneous kernel; Lean derives both its unique minimum-norm property and the scaled decomposition (MinimumNormConstraintBase.norm_le_of_maps_to_target, MinimumNormConstraintBase.unit_candidate_scale_le). The formalization also constructs this certificate. For linearly independent environment vectors $\mu_1, \ldots, \mu_E$ with Gram matrix G, and any nonzero target t, it defines

$$q = \sum_i (G^{-1}t)_i \, \mu_i.$$

The constraint map $x \mapsto (\langle \mu_i, x\rangle)_i$ sends q to t, and every homogeneous residual is orthogonal to q (environmentConstraintMap_gramMinimumNormBase, linearlyIndependent_environment_minimumNormConstraintBase). Thus the normalized scale bound and its equality case follow constructively from independence, rather than from an assumed minimum-norm vector (linearlyIndependent_environment_unit_candidate_scale_le, linearlyIndependent_environment_unit_candidate_unique). Specializing $t_i = \sigma_i^2 > 0$ gives the geometric statement of Lemma C.1 directly: the normalized direction has unit norm, satisfies every $p^\top \mu_i = \widetilde{\mu}\sigma_i^2$ equation, maximizes $\widetilde{\mu} \geq 0$, and is unique at the maximum (linearlyIndependent_positiveVariance_unique_maximalDirection). A repeated-vector family has singular Gram matrix, recording the required negative boundary (repeatedPlaneEnvironments_gram_not_isUnit). A two-dimensional first-coordinate constraint with base norm two has a unique unit candidate at maximal scale $1/2$; without orthogonality, scalar cancellation admits scale two and refutes the bound. These results do not yet prove the equality-dimensional spurious construction or formalize the paper's Gaussian, logistic-risk, and zero-one-risk comparisons.

Localized tangent features give an exact noninterference criterion. Ortiz-Jimenez, Favero, and Frossard [50] express task t in a kernel eigenbasis as $f_t(x) = \sum_\rho c_{t,\rho}\phi_\rho(x)$. Their

Proposition 2 says that task arithmetic is exact when every feature carrying a nonzero coefficient for task t is supported inside its domain D_t. The finite formalization strengthens the all-ones sum to every mixing vector α:

$$x \in D_t \quad \Longrightarrow \quad \sum_u \alpha_u f_u(x) = \alpha_t f_t(x).$$

This weighted property is equivalent to the observable boundary that f_u vanishes on D_t whenever $u \neq t$ (`weightedTaskArithmetic_iff_noCrossTaskInterference`). Pairwise-disjoint domains and active-feature localization imply it directly (`localizedActiveFeatures_weightedTaskArithmetic`).

The converse records the load-bearing assumption rather than hiding it. On each domain, delete features whose restriction is identically zero and require the remaining restrictions to be linearly independent. Lean proves that this ordinary linear independence yields coefficient separation (`locallyIndependentOrZero_localCoefficientSeparation`). If the disjoint task domains cover the point type, weighted task arithmetic is then equivalent to localization of every active feature (`weightedTaskArithmetic_iff_activeFeaturesLocalized`), the finite scalar counterpart of Proposition 3. Coordinate features on two singleton domains satisfy the equivalence. Conversely, two individually nonlocalized features can cancel on the other domain and still give exact task arithmetic; that representation is proved not locally separated (`cancellingRepresentation_taskArithmetic_not_localized`, `cancellingFeatures_not_locallySeparated`). Removing the cancelling coefficient yields explicit interference (`nonlocalized_without_cancellation_fails_taskArithmetic`). The theorem concerns finite exact representations; it does not identify trained neural features with NTK eigenfunctions, prove convergence of an infinite RKHS expansion, or transfer exact arithmetic to finite-precision runtime behavior.

Detached superfluous outputs recover exactly the intended gradient. SupSup [51] combines task masks before evaluating a fixed random network. At a fixed-feature linear head, this construction has an exact algebraic boundary. For arbitrary real coefficients α_t and masks M_t,

$$\ell\left(\sum_t \alpha_t M_t\right) = \sum_t \alpha_t \ell(M_t),$$

and a one-hot coefficient recovers the selected task mask exactly (`maskedLinearLogit_weightedSupermask`, `weightedSupermask_oneHot`). A nonlinear activation cannot in general be moved across this sum: a scalar rectifier gives a concrete cancellation counterexample (`nonlinear_activation_breaks_task_logit_superposition`).

Appendix I of the source partitions the output coordinates into real classes R and added superfluous classes S. It detaches R and differentiates log-sum-exp. If the supervised target c lies in R, the resulting declared gradient is zero on R and, for every $v \in S$, is exactly

$$\frac{e^{y_v}}{\sum_j e^{y_j}} = \big(\nabla_y \, \mathrm{CE}(c, y)\big)_v.$$

Lean proves both coordinate statements and the vector identity saying that the detached objective's gradient is precisely the supervised softmax-minus-one-hot gradient restricted to S (`detachedLogSumExpGradient_eq_crossEntropyGradient_of_mem`, `detachedLogSumExpGradient_eq_zero_of_not_mem`, `detachedLogSumExpGradient_eq_restrict_crossEntropyGradient`). The target partition is essential: if c is incorrectly placed in S, the two gradients differ by exactly one at c (`target_superfluous_gradient_difference_eq_one`). These results recover Equation 19's linear head and Lemma I.1; they do not establish one-shot task recovery, the later expectation-sign claims, learned mask quality, Hopfield storage, or the empirical results.

Top-N prompt query has an exact exchange boundary. Learning to Prompt (L2P) [52] chooses N distinct prompt keys by minimizing the sum of their query–key matching costs. For a finite pool P, cost c, and feasible selection $S \subseteq P$ with $|S| = N$, Lean proves the exact characterization

$$S \in \arg\min_{\substack{T \subseteq P \\ |T|=N}} \sum_{i \in T} c_i \quad \iff \quad \forall i \in S,\ \forall j \in P \setminus S,\ c_i \leq c_j.$$

The forward direction swaps a selected and rejected prompt. The reverse direction pairs the equal-cardinality set differences $S \setminus T$ and $T \setminus S$, so the result includes ties and arbitrary finite prompt types (`minimizer_noCrossBoundaryInversion`, `noCrossBoundaryInversion_minimizer`, `minimizesSelectionCost_iff_noCrossBoundaryInversion`).

L2P's optional training-time diversity rule replaces c_i by $c_i h_i$, where h_i is historical selection frequency. A uniform $h_i = h > 0$ preserves the complete set of minimizers (`minimizesSelectionCost_uniformFrequency_iff`). Nonuniform frequency is genuinely different: in a two-prompt fixture the raw costs are $1 < 2$, but the strictly positive frequencies $(3, 1)$ change the weighted costs to $(3, 2)$. Prompt zero is the raw-cost minimizer, prompt one is the frequency-weighted minimizer, and prompt zero is no longer a weighted minimizer (`firstPrompt_raw_minimizer`, `secondPrompt_positiveFrequency_minimizer`, `firstPrompt_not_positiveFrequency_minimizer`, `positiveReuseFrequency_positive`). This finite result recovers Equations 3–4; it does not formalize prompt concatenation, learned query/key maps, Equation 5's surrogate loss, task inference, or empirical retention.

Prefix prompting preserves query arity, not token output. DualPrompt [53] distinguishes prompt tuning, which prepends the prompt to queries, keys and values, from prefix tuning, which prepends it only to keys and values. For $p > 0$ prompt tokens and L input tokens, their output-query counts are respectively $p + L$ and L (`promptTuningQueryCount_eq_prefixTuningQueryCount_iff`, `prefixTuning_preserves_queryCount_promptTuning_increases`).

Preserving query count does not imply preserving the old token values. In one scalar attention coordinate, let $M_P, M_X > 0$ be the unnormalized softmax masses of a nonempty prompt and token context, with separately normalized outputs a_P, a_X. Lean proves the exact displacement

$$a_{P+\!\!+X} - a_X = \frac{M_P}{M_P + M_X}(a_P - a_X)$$

and hence $a_{P+\!\!+X} = a_X \iff a_P = a_X$ (`scalarPrefixAttention_sub_tokenAttention`, `scalarPrefixAttention_eq_tokenAttention_iff`). Equal-score scalar fixtures realize both strict change, from 1 to 2, and exact neutrality (`prefix_changes`, `prefix_neutral`). This recovers Equations 4–5's arity distinction and supplies its exact attention boundary; it does not establish learned prompt quality, task-key identification, rehearsal freedom or benchmark accuracy.

Hierarchical prompt loss requires an inequality, not budget substitution. HiDe-Prompt [55] factors correct-class probability into within-task prediction and task-identity inference. For positive probabilities $p_{\mathrm{WTP}}, p_{\mathrm{TII}}$, negative-log loss makes the factorization exact:

$$-\log(p_{\mathrm{WTP}} p_{\mathrm{TII}}) = -\log p_{\mathrm{WTP}} - \log p_{\mathrm{TII}}$$

(`negativeLogLoss_mul`). Writing the three nonnegative component losses as $H_{\mathrm{WTP}}, H_{\mathrm{TII}}, H_{\mathrm{TAP}}$, the complete loss is

$$L = \max\{H_{\mathrm{WTP}} + H_{\mathrm{TII}}, H_{\mathrm{TAP}}\}.$$

Component bounds δ, ϵ, η therefore imply $0 \le L \le \max\{\delta+\epsilon, \eta\}$, and conversely $L \le \xi$ forces each component below ξ (`hierarchicalLoss_mem_Icc_of_component_bounds`, `componentLosses_le_of_hierarchicalLoss_le`).

The source theorem states the interval correctly, but its appendix substitutes the upper bounds as exact losses. That step is invalid: all three losses may be zero while $(\delta, \epsilon, \eta) = (1, 1, 1)$, giving $L = 0$ but envelope 2 (`component_upper_bounds_do_not_force_envelope_equality`). Moreover, equal joint probability can hide different allocations between within-task and task-identity performance, and strict improvement in one factor is visible only when the other has positive mass (`equal_joint_probability_different_components`, `zero_task_probability_masks_within_improvement`). These results audit the source's probability and loss algebra; they do not establish its learned prompts, pseudo-feature distributions, head optimization, rehearsal freedom or empirical accuracy.

Decomposed prompting separates coefficient retention from behavior retention. CODA-Prompt [54] replaces discrete prompt selection by the input-conditioned linear combination

$$p(x) = \sum_m \alpha_m(x) P_m.$$

Because each α_m is computed from that component's own key and attention vector, appending components preserves the complete prefix of old weights exactly (`componentWeights_take_old_after_append`). The assembled prompt, however, obeys

$$p_{\text{old++fresh}}(x) = p_{\text{old}}(x) + p_{\text{fresh}}(x),$$

and therefore equals the old prompt exactly when the fresh weighted contribution vanishes (`assembledPrompt_append`, `assembledPrompt_append_eq_old_iff`). A scalar fixture retains the old coefficient while changing the assembled value from 2 to 14, so frozen old weights are not a no-regression theorem (`scalar_expansion_preserves_old_weights_but_changes_prompt`). If the coefficients were instead globally softmax-normalized, any finite new logit would reduce every positive old route weight (`finiteSoftmaxExpansion_strictly_decreases`, `globallyNormalized_expansion_changes_old_weight`).

The source also penalizes $\|BB^\top - I\|$. Its squared-Frobenius version has the same zero set, and Lean proves that the penalty vanishes exactly when the rows of B are orthonormal (`codaOrthogonalityPenalty_eq_zero_iff_rowsOrthonormal`). Identity rows give zero, while repeating one unit row gives squared penalty two (`identity_penalty_zero`, `repeated_row_penalty_eq_two`). These are finite component and Gram identities; they do not establish the quality of learned similarities, end-to-end transformer optimization, forgetting, privacy, or accuracy.

Checkpoint averaging has an exact affine boundary. The model-soup recipe of Wortsman et al. [56] averages independently fine-tuned parameters, while its greedy variant retains an ingredient only when validation performance does not decrease. For an affine predictor $f(\theta) = A\theta + b$, parameter interpolation and output ensembling agree exactly:

$$f((1-\alpha)\theta_0 + \alpha\theta_1) = (1-\alpha) f(\theta_0) + \alpha f(\theta_1)$$

(`affinePredict_twoSoup`). The finite greedy construction has a separate, assumption-free invariant: every accepted-or-rejected proposal leaves the validation score of the current soup nondecreasing, so a best-first initialization cannot finish worse than its first ingredient on that same validation score (`soupScore_greedySoup_ge`, `greedySoup_no_worse_than_first`).

This does not license averaging arbitrary BP, PC, or carrier checkpoints. For the nonlinear scalar predictor $f(\theta) = \theta^2$, the endpoints $\theta = -1$ and $\theta = 1$ both predict one, but their uniform parameter soup predicts zero while their output ensemble predicts one; against target one, the soup is strictly worse than both endpoints (`nonlinearSoup_ne_ensemble`, `uniformSoup_can_be_worse_than_both_endpoints`). Any checkpoint soup therefore needs coordinate identity, validation provenance, path-loss and legal-policy measurements rather than an appeal to endpoint quality.

Executable boundary. The exact-real transition has a recorded addressed-packet witness whose child order, type indices, binary32 words and nonzero fresh packet are checked together (`productionInvocation_certificate`). A separate decision-site fixture decodes real binary32 weights and observations and proves all coordinate errors within their budgets (`certificateBatch_total_error_is_bounded`). These are authenticated pointwise witnesses, not a claim that every tensor operation in a training run has been reconstructed in the kernel.

The registered PC-from-update-zero instance uses 60 slots of content width 112, evidence width 16, recurrent-control width 48, and four routed operators. It has 5,359,248 parameters against the matched recurrent carrier's 5,279,800, while 581,400 trainable scalars occupy 66 tensors. Persistent cross-episode evidence is disabled, so that campaign tests the within-episode carrier rather than the full persistence hypothesis. Terminal efficacy claims are admitted only after the generation-eleven–thirteen independent seal.

The mechanism trace separates direction from magnitude. Raw-credit cosine describes the local credit before optimizer transport, whereas optimizer-displacement cosine describes the transported parameter direction; neither bounds the size of that motion. The formal telemetry boundary exhibits both cosines equal to one with transported norm ratio 10^{-3} (`cosine_eq_one_not_normRatio_le`). The terminal analysis therefore reports both cosine and norm ratio on both sides of optimizer transport, while verified yield and work remain separate empirical quantities.

The comparison crosses recurrent, typed-workspace and unified-routed carriers with ordinary BP, an independent BP opportunity control, prospective PC, multi-site error-coordinate PC and incremental PC. All five rules begin from identical bytes within a carrier; no rule-specific generation-ten checkpoint or BP warm-up is used. PC is active at the first update, although the registered admission gate permits BP-equivalent ignition and measures later departure rather than requiring instant distinctness. The branch[1] retains all 15 cells through its prospectively fixed three-round boundary, in one seeded world.

6.1 Recorded branch tables (mis-specified continuation; see footnote above)

Verified union 747; the independent-BP control is the largest single rule (400), incremental PC fails (56); yields decline every round (a resource-boundary observation, not a saturation proof: `not_correct_on_zeroTail_and_oneTail`). By rule and by carrier:

[1] A botched experiment that nonetheless adds data. It was intended as a from-scratch comparison of update rules; it was mis-specified and executed as a continuation from generation-eight weights (the last rule-neutral common ancestor, trained by the earlier backpropagation loop) over the generation-ten single-world pool. Its verified discoveries enter the solution pool with full provenance; its rule and carrier comparisons are recorded for completeness and are not evidence on the from-scratch question. Row labels 11–13 are pool-accounting indices, not scientific generation numbers; the branch is parallel to, not a successor of, the generation-nine/ten factorial.

rule	round 1	round 2	round 3	cumulative	exclusive	train h
bp_control	198	118	62	378	7	4.227
bp_independent	234	102	64	400	287	4.133
incremental_pc	27	12	17	56	43	39.159
multi_site_error_coordinate_pc	199	118	64	381	17	12.857
prospective_pc	202	115	66	383	7	5.242

carrier	round 1	round 2	round 3	cumulative	exclusive	train h
gru	200	84	59	343	271	7.469
typed_workspace	85	90	37	212	139	22.481
unified_routed	186	93	55	334	208	35.668

The final round is shown cell by cell; optimizer cosine and norm ratio are reported together, and retention NLL remains a separate endpoint. The retention column measures teacher-forced, fixed-budget behavioural retention only—it bounds neither cross-task transfer nor what an unconstrained search would recover:

pool idx	carrier	rule	new	excl.	train h	opt. cos.	opt. norm	retention NLL
13	gru	bp_control	31	0	0.205	1.0000	1.000000	0.2738
13	gru	bp_independent	30	22	0.205	1.0000	1.000000	0.2603
13	gru	incremental_pc	1	0	1.603	0.9891	0.003951	0.4631
13	gru	multi_site_error_coordinate_pc	31	3	0.280	1.0000	0.999911	0.2737
13	gru	prospective_pc	32	0	0.226	1.0000	0.999911	0.2738
13	typed_workspace	bp_control	22	0	0.469	1.0000	1.000000	0.3730
13	typed_workspace	bp_independent	14	10	0.629	1.0000	1.000000	0.3651
13	typed_workspace	incremental_pc	2	2	4.721	0.9886	0.001969	0.5211
13	typed_workspace	multi_site_error_coordinate_pc	22	1	1.459	1.0000	0.999940	0.3726
13	typed_workspace	prospective_pc	23	0	0.686	1.0000	0.999944	0.3728
13	unified_routed	bp_control	19	0	0.937	1.0000	1.000000	0.3721
13	unified_routed	bp_independent	21	16	0.691	1.0000	1.000000	0.3643
13	unified_routed	incremental_pc	14	12	7.799	0.9964	0.000980	1.3931
13	unified_routed	multi_site_error_coordinate_pc	16	1	3.084	1.0000	0.999937	0.3714
13	unified_routed	prospective_pc	21	3	1.223	1.0000	0.999941	0.3723

Because persistent cross-episode evidence was disabled, these rows adjudicate the within-episode unified carrier only. They do not adjudicate the persistent evidence plane or any proposed single-variable ablation.

Opportunity control: an independent BP seed given identical opportunity weakly dominates all nine candidate arms on the count/work Pareto:

carrier	rule	ext.	indep. BP ext.	beyond 2 BP	work ×BP	BP dom.
gru	incremental_pc	10	130	10	7.445	yes
gru	multi_site_error_coordinate_pc	8	130	8	1.330	yes
gru	prospective_pc	1	130	0	1.073	yes
typed_workspace	incremental_pc	4	94	4	8.945	yes
typed_workspace	multi_site_error_coordinate_pc	11	94	11	2.845	yes
typed_workspace	prospective_pc	5	94	5	1.159	yes
unified_routed	incremental_pc	36	156	34	10.524	yes
unified_routed	multi_site_error_coordinate_pc	9	156	9	3.879	yes
unified_routed	prospective_pc	11	156	11	1.414	yes

Within the branch the unified-routed carrier is the strongest carrier for every PC rule (largest beyond-two-BP margin 34; prospective at 1.41× work); the selection inequality of Section 12 funds BP seeds first on these counts.

The remaining decisive experiments are single-variable: recurrent control on/off; precision-read bias on/off; Bayes gate versus unit gain; persistent prior on/off; and the carrier crossed with a mechanism-distinct credit rule. All share the unchanged typed-action waist and external checker.

7 The plasticity axis: what of predictive coding is a backprop trick

Predictive coding is often presented as a rival to backpropagation. Within the finite, declared family of mechanisms we model, most of it is a preconditioner.

Theorem 7.1 (Subsumption partition)**.** *Of six named mechanisms, five have constructive backpropagation realizations—an explicit preconditioned or proximal update reaching the same endpoint or bound—and the sixth, synchronized local parallelism, is an* execution-model *property rather than a training-rule advantage (`bp_subsumption_partition_crown`). Unit-rate error-coordinate predictive coding simply is the backpropagation product (`identityPreconditionedPC_bpEquivalentLicense`, `epcOneStep_bpEquivalentLicense`), and curvature-normalized preconditioning is loss-decreasing exactly on the open interval* $(0, 2)$*, with unit rate the one-step optimum.*

Two caveats are load-bearing. Larger negative curvature is not an escape algorithm: at an exact saddle a preconditioned gradient step still has zero gradient, so escape requires a perturbation or an explicit negative-curvature step, and the partition states this rather than eliding it. And the execution-model residue is not a free win either—on a unit-bandwidth synchronous chain, exact predictive coding has no strict latency advantage (`exactUnitBandwidthPC_no_strictDigitalLatencyAdvantage`), while propagation remains bounded by topology distance and local bandwidth (`pcFiniteSpeed_distance_le_sweeps_mul_bandwidth`). The genuine residue is the serial-versus-parallel gap (`synchronizedLocalCost_serial_parallel_exact`), which is a statement about hardware, not about learning rules.

The continual-learning claim deserves the same treatment. Under matched target progress, restriction, exact scheduling and vanishing interference, a predictive-coding adapter can still forget *more* than a backpropagation adapter—the two solutions are reflections across the optimum (`frozenAdapter_four_hypotheses_insufficient`). The advantage is restored only by a fifth condition requiring both residuals to share a sign (`frozenAdapter_five_hypotheses_restore_sourceForgetting_bound`). That condition is a controlled-adaptation assumption, and uncontrolled online data is precisely where it fails.

8 Depth is repairable; cost is not

Iterative relaxation degrades badly with depth, and the degradation has a diagnosis: the per-layer precision-weighted energy is imbalanced across the stack, and a stability-bounded activity rate slows the error front so that layer l is reached only after $L - l$ sweeps. The PCX benchmark of Pinchetti et al. [45] makes this an executable measurement programme: compare the PC variants against backpropagation on shared architectures while recording layer energy, stability, and wall time rather than accuracy alone. Recent work repairs this to accuracy comparable with backpropagation on deep residual networks, and we have formalized the mechanisms at arbitrary depth over full covariance matrices with a nonlinear Jacobian remainder: the first-arrival attenuation that produces the imbalance, a precision spike that exactly cancels it, a forward-reference update that bounds deep-layer accumulation, auxiliary buffers that synchronize residual and main paths, and the invariance of frozen normalization statistics (`vectorDepthRepair_crown`, `depthScalingRepair_crown`).

The scope should be read narrowly. These are exact local identities and propagation bounds. Convergence and accuracy for trained deep networks are classified in the development as empirical targets, not theorems. And no repair touches the price: the repaired dynamics remain an iterative relaxation, so "matches backpropagation" continues to mean matches its target, not its cost (`pcBPCompute_crown`).

9 Curvature spread decides both the signal and the solver

Depth is one way an iterative relaxation degrades. Curvature spread across the sites being settled is another, and it turns out to govern two things at once—how far the settled credit can depart from backpropagation, and whether the settling procedure can reach it at all. On a block-diagonal quadratic settle energy, which is the exact local model of multi-site settling near an equilibrium, the following are proved.

Solver substitution cannot move an equilibrium. A single global rate, per-block rates, and Levenberg-style damping have identical stationary sets (`stationary_set_shared`). Changing how one settles therefore changes the path and not the destination. This is an equilibrium statement only: it does not by itself license a solver swap at the finite settling depth used by a trainer.

The finite-depth license can now be stated exactly. Let F and G be contractions with factors $q_F, q_G \in [0, 1)$ and a shared fixed point z. For initial states x_F, x_G, settling depths T_F, T_G, and a credit readout R that is K-Lipschitz about z, the machine-checked bound is

$$\left\|R(G^{T_G}x_G) - R(F^{T_F}x_F)\right\| \leq K\left(q_G^{T_G}\|x_G - z\| + q_F^{T_F}\|x_F - z\|\right).$$

Thus solvers, initializers, and work budgets may all differ, but every difference is charged to the finite-credit budget (`finiteSolver_credit_difference_le`). If that budget is below a registered tolerance, substitution is admitted (`finiteSolver_credit_difference_le_tolerance`); if it is smaller than the observed candidate-credit norm, the candidate remains positively aligned with the reference finite credit (`finiteSolver_positiveAlignment`). This is the finite-T statement required by an actual boundary amendment. The nonlinear version makes the basin premise explicit: each solver supplies an invariant local-contraction neighborhood containing the shared target and its own initializer (`localFiniteSolver_credit_difference_le`).

Warm starting has the sharper same-solver bound

$$\left\|R(F^T x_{\text{warm}}) - R(F^T x_{\text{cold}})\right\| \leq K q^T \|x_{\text{warm}} - x_{\text{cold}}\|.$$

Moreover, if the warm initializer is already q^s times closer to the fixed point, then T warm-started steps have no larger certified credit-error budget than $T + s$ cold-started steps (`warmStart_credit_difference_le`, `warmInitializer_savesSteps_creditBudget`). This licenses a reduction in inference work only when the initializer-distance certificate is supplied; state reuse alone is not the certificate.

The comparison can be made sharper still for the finite endpoint that a trainer actually registers. Suppose the warm proposal runs for m steps while the cold reference runs for $m+s$ steps under the same q-contractive solver. Pairwise K-Lipschitz continuity of the readout gives the fixed-point-free bound

$$\left\|R(F^m x_{\text{warm}}) - R(F^{m+s} x_{\text{cold}})\right\| \leq K q^m \left(\|x_{\text{warm}} - x_{\text{cold}}\| + \left(\sum_{j=0}^{s-1} q^j\right)\|x_{\text{cold}} - F x_{\text{cold}}\|\right).$$

The first term transports the initializer mismatch; the second charges exactly the cold tail omitted by early stopping, using its observable first-step residual. No exact fixed point and no claim of stationarity is needed (`warmShort_credit_difference_to_coldLong_le`). For a nonlinear solver, the same statement requires the warm and cold initializers to inhabit one invariant local-contraction neighborhood (`local_warmShort_credit_difference_to_coldLong_le`). This is the appropriate same-basin obligation for comparing a short warm solve directly with the registered longer cold solve.

Execution adds a second boundary. If the realized warm and cold credits lie within ϵ_w and ϵ_c of their exact-real readouts, respectively, then the complete realized-credit error is at most

$$\epsilon_w + Kq^m \left(\|x_{\text{warm}} - x_{\text{cold}}\| + \left(\sum_{j=0}^{s-1} q^j \right) \|x_{\text{cold}} - Fx_{\text{cold}}\| \right) + \epsilon_c.$$

This composition is machine checked (`local_realized_warmShort_credit_difference_to_coldLong_le`); the precision terms must be supplied by endpoint replay certificates rather than inferred from solver telemetry.

The solver map itself must also match the implementation. In the registered multi-site settle, the error gradient is divided by row masses before stepping and the proposed rate is selected by a finite monotone-energy backtracking search. Let P be the fixed inverse-mass linear map, J_0 the raw gradient Jacobian at an audited center, and suppose the raw Jacobian varies by at most δ throughout the declared ball. If PJ_0 has strong-monotonicity modulus μ, then preconditioning spends at most $\|P\|\delta$ of that margin. The exact regional conditions are

$$\|P\|\delta < \mu, \qquad \eta(\|PJ_0\| + \|P\|\delta)^2 < 2(\mu - \|P\|\delta),$$

together with the audited center-displacement admission bound. Under them the fixed proposal $x \mapsto x - \eta P \nabla E(x)$ supplies the local contraction consumed above (`PreconditionedJacobianEnclosure.toLocalContractionCertificate`).

That fixed-map certificate transfers to the executable backtracking map only if the initial proposal passes the monotone-energy predicate at every point of the admitted ball. This branch-stability obligation is explicit and machine checked (`PreconditionedJacobianEnclosure.toMonotoneBacktrackingCertificate`). It cannot be replaced by "no backtracks were observed" at a few telemetry points. For the scalar square energy, rate 3 sends 1 to -2 and is rejected, while one half-rate retry sends 1 to $-1/2$ and is accepted; hence backtracking and the nominal fixed-rate map genuinely differ when the initial branch is not stable (`monotoneBacktracking_rejects_unstable_initial`). Exact-real energies are finite by construction; a floating-point realization must additionally reject non-finite values and bind its endpoint to the separate precision budget.

For the block-quadratic model the comparison is not merely bounded. On block i, the global and damped finite states differ exactly by

$$\left[(1 - \eta\kappa_i)^T - \left(\frac{\lambda}{\kappa_i + \lambda} \right)^T \right] (x_i - z_i),$$

and a linear credit readout multiplies this expression by its block weight (`global_damped_iterate_difference`, `coordinateCredit_global_damped_difference`). The hypotheses are necessary: the half-step and quarter-step scalar solvers have exactly the same stationary set but different one-step credit (`shared_stationary_set_not_finiteStep_license`).

Finite-credit proximity is still not the scientific objective. The accelerated credit must preserve whatever conditional advantage justified the cold predictive direction. Let d_c be the authenticated cold finite credit, d_w the candidate warm credit, b the comparison direction, and g the retention-weighted gradient. If $\|d_w - d_c\| \le \varepsilon$, then the cold advantage survives whenever

$$\|g\|\varepsilon < \langle g, d_c \rangle - \langle g, b \rangle.$$

For finite parameter steps the gate additionally charges the difference between the candidate and baseline directional-curvature bounds. Both versions are machine checked (`finiteSolver_preserves_retentionAdvantage`, `finiteSolver_preserves_decreaseLowerAdvantage`). A fail-closed selector uses the candidate only on admission and otherwise returns the cold reference exactly; its advantage theorem requires the candidate gate only on the admitted branch (`guardedFiniteSolver_preserves_retentionAdvantage`). The finite-trajectory form composes the same-basin, omitted-tail, and two endpoint precision budgets directly with this fail-closed advantage gate (`guardedLocalWarmShort_preserves_retentionAdvantage`). Recomputing the complete cold endpoint merely to evaluate the admitted branch can erase the proposed saving. A sharper form therefore accepts a certified lower bound M on the exact-real cold endpoint's advantage. If the warm trajectory and its one-sided replay budget give $\|d_w - d_c\| \le \varepsilon$ and $\|g\|\varepsilon < M$, then the warm candidate is admitted without executing d_c. Rejection still computes and returns an authenticated cold fallback (`guardedLocalWarmShort_withoutAdmittedColdReplay`). The lower bound M is a real certificate obligation; historical average advantage or a sampled cold score cannot replace it.

Finally, fewer sweeps are not yet less work. Write

$$W_p = W_{\text{init}} + W_{\text{cert}} + W_{\text{fp}} + W_{\text{audit}} + cT_w, \qquad W_c = W_{\text{cold,fixed}} + cT_c$$

for proposal and cold-reference work. An admitted proposal accelerates exactly when $W_p < W_c$; a rejected proposal costs $W_p + W_c$ and therefore cannot accelerate that update. Across A admissions and R rejections, the exact threshold is

$$(A+R)W_p + RW_c < (A+R)W_c \quad \Longleftrightarrow \quad (A+R)W_p < AW_c$$

(`admitted_guardedWork_lt_coldWork_iff`, `rejected_guardedWork_not_lt_coldWork`, `batchGuardedWork_lt_coldWork_iff`). In the executable fixture, four saved sweeps remain beneficial with two units of certification overhead, but the same sweep saving loses once overhead reaches five; with one rejection, three admissions merely break even and four are needed for strict net saving. Thus safe fallback and acceleration are separate claims: the former protects the experiment, while only complete work accounting establishes the latter.

A single global rate is defeated by spread. Any rate above $2/L$ for the stiffest block amplifies that block's error at every step (`stiff_block_unstable`), while any rate small enough to be stable there retains at least $1 - T \cdot (2\tau/\sigma)$ of a tame block's error after T steps (`tame_block_starved`); with curvatures $\{1, 10^6\}$ and eight steps the tame block keeps at least 99% of its error. Per-block rates c/curvature instead contract every block by the same factor $1 - c$ regardless of spread (`blockStep_spread_invariant`), and damping contracts unconditionally, its per-block factor lying in $[0, 1)$ for every damping parameter (`dampedStep_factor_stable`). Blockwise and damped stepping are thus the solver family whose stability does not degrade as spread grows.

A residual tolerance certifies nothing about credit. Credit error is bounded by the credit-weight bound times the residual, but no uniform constant exists: for every gradient tolerance and every demanded credit-error bound there is a model violating it (`residual_tolerance_insufficient_for_credit`). A settling criterion stated in gradient units therefore has no consequence for the quantity actually being transported, and must be stated in credit units or accompanied by a per-model bound.

Departure needs spread, not degeneracy. At curvature ratio two, with every block strictly contracting under the global rate, the settled credit has positive inner product with the instantaneous credit, is not a scalar multiple of it, and has squared cosine strictly below one—by exact rational arithmetic (`bounded_spread_departure`). The regime in which predictive coding is genuinely non-backpropagation is thus reachable at bounded conditioning; it is not an artifact of the numerics degenerating.

Read together these say something sharper than any one of them. The same quantity that makes settled credit interesting—spread—is what defeats the naive solver, so the interesting regime and the badly-conditioned regime are adjacent but not identical, and the correct response to instability there is a metric change rather than a smaller step. What remains open is quantitative: a bound relating spread to attainable cosine, and a prediction of the depth at which transported credit saturates. The measurements in the companion experiments show that saturation occurring far earlier than energy convergence, which the block-diagonal model does not yet explain.

10 Forgetting has a metric side and a connection side

Sequential learning departs from additive learning in two independent ways. For two quadratic tasks with curvatures A_1, A_2, step size η and gradient g_1,

$$\theta_{1\to 2} - \theta_{\text{add}} = \eta^2 A_2 g_1(\theta), \qquad \theta_{1\to 2} - \theta_{2\to 1} = \eta^2 [A_1, A_2]\theta.$$

The first is a *metric* effect—double counting, present even when the order cannot matter. The second is a *connection* effect: order matters exactly to the extent that curvatures fail to commute. A rank-one trichotomy separates them cleanly: parallel tasks commute and any forgetting is pure double counting; orthogonal tasks do not interact; oblique tasks carry the irreducible commutator (`rankOne_parallel_orthogonal_oblique_trichotomy`). Per-pair conflict is measured exactly by the diagonal of an interference Gram built from commutators, which vanishes iff the pair commutes (`pairwiseInterferenceEnergy_eq_zero_iff_commute`).

Two natural-looking claims are false, and the counterexamples are machine-checked. A trivial rotational factor does *not* imply commutation in general: a palindromic product ABA of symmetric matrices stays symmetric when $[A, B] \neq 0$. And interference *spectra* transfer under any similarity (`matrixCommutator_charpoly_unitConjugate`) while interference *angles* transfer only under orthogonal conjugation—an explicit shear breaks them. Pointwise agreement of values and gradients does not determine curvature at all (`pointwiseAgreement_not_determineHessian_negativeExample`), so behavioural similarity is never sufficient to trust a transferred measurement.

The unifying observation is that one Gram matrix does double duty: it certifies both a curriculum's right to be reordered and a routed command schedule's right to be treated as a set (§5). The continual-learning side and the architecture side are the same fact seen from two directions.

11 Fast state versus weights

A system may adapt at inference time in bounded fast state—a context, a cache, a bank of evidence registers—over frozen parameters, and periodically consolidate that state into weights. This is the deployment question behind personalization: how much adaptation belongs in state, and how often, if ever, must weights move?

The results in this section are theorems about a linear-effect model: the readout is the sum of a slow effect and a fast effect, each a linear map of its argument. We state the commitment here rather than in a limitation, because it is doing the work.

Theorem 11.1 (Faithfulness is reachability)**.** *In a linear-effect model, consolidating fast state into weights preserves the adapted readout exactly when the slow update reproduces the fast effect (`faithfulConsolidation_iff_effect_eq`); some faithful consolidation exists exactly when the fast effect lies in the range of the weight-update map (`exists_faithfulConsolidation_iff_mem_range`). A transverse fixture outside that range has an exact irreducible consolidation loss.*

Theorem 11.2 (Capacity forces a deadline)**.** *Under constant arrivals at rate r into capacity C, all evidence is admitted before the first saturation time $\lceil C/r \rceil$, and from that point admitted evidence is exactly C while loss is exactly $rt - C$. The deadline is computed, not chosen.*

Three further boundaries follow. Consolidation double-counts unless the reused contribution is discounted: carrying it at full weight yields exactly one extra copy in both precision and natural-parameter coordinates, and zeroing that contribution restores once-each calibration (`reused_contribution_exact_excess`). The order of consolidations matters exactly when their curvatures fail to commute—the same Gram criterion as §10—so with oblique periods, permuting two of them yields a different final model. And in a stationary, within-capacity, fast-reachable regime, pure fast-state adaptation attains zero prediction error while any equally faithful consolidation attains the same error at strictly greater cost; consolidation becomes necessary exactly when the weights can realize the shift *and* either capacity is exceeded or the shift leaves fast-state reachability (`twoTimescaleAdaptation_crown`).

Two families of practice bracket this section. On the weight side, constrained-update continual learning already builds the projections our faithfulness condition characterises: episodic-memory methods constrain an update against remembered losses [1], and orthogonal-gradient methods project into a subspace leaving earlier outputs unchanged [2]—both are instances of the minimum-change update our reachability criterion licenses, and both inherit its incompatibility boundary. On the state side, a growing family makes the fast state itself the adapting object: test-time-training layers treat the hidden state as a model updated by self-supervised learning [27], one recurrence is derived directly from implicit online regression [28], and gated delta rules add learned erasure to targeted writes [29]. Our additive evidence registers are the conservative, provably calibrated member of that family; the gated variants trade exact fusion for the forgetting a drifting world requires, which is precisely the boundary of Section 4.

That last statement is the one with practical force: it says weight updates are not the default, they are what you resort to when bounded state provably cannot carry the adaptation. How often to consolidate inside the feasible band is now derived rather than declared: tracing the objective to the development's own accounting—a counted update cost falling like $1/T$ against the hinge of lost evidence, identically zero before the saturation deadline—the unique positive-period optimum is the *deadline itself* whenever losing a full register costs more than an update (`derivedCadenceObjective_at_exactDeadline`). An earlier draft of this theory optimized a posited symmetric quadratic instead; it selects an interior period, disagrees with the derived objective by an exact computed gap on the same fixture, and survives only as a declared approximation. Consolidate at the deadline the capacity computes, not on the schedule a smooth objective manufactures.

12 Search under a budget, and the corpus it produces

A discovery system spends a fixed budget of draws from a learned prior and keeps what an external checker verifies. Three measured facts from our own campaigns had no theory until recently: roughly three quarters of all draws were duplicates of programs already proposed; the four architectures' discoveries were 8–16% overlapping, with about 70% of the union found by exactly one architecture; and a fixed-split portfolio underperformed its own best member. Each is now a theorem about the draw model rather than an anecdote.

Expected *distinct* coverage under n independent draws is $\sum_t (1 - (1 - p_t)^n)$. The exact identity is `iidExpectedDistinctCoverage_eq_sum_one_sub_pow`; the duplicate burden is exactly expected hits minus expected distinct coverage (`iidExpectedDuplicateBurden_eq_hits_sub_distinct`); the two-draw collision probability is the prior's collision mass (`twoDrawCollisionProbability_eq_collisionMass`), which is what a 75%-duplicate search is measuring about its own prior. Deduplicating draws by a semantics-preserving canonical form buys back exactly the collapsed collision mass—the formal price tag on canonicalization, computed before building it. Sharpness has an interior optimum determined by ranking quality, not entropy alone, and two priors with equal entropy and different gold mass have strictly ordered yield: decisiveness is not a dial that buys ranking. The portfolio results close the loop (`budgetedSearchYield_crown`): equal standalone yields can carry radically different union value through complementarity, and a fixed split can lose to the best single arm while an adaptive allocation dominates both—which is the observed 399-versus-435 in one line.

Coverage alone still omits the cost of the arm that produced it. Let

$$U(P) = v\left|\bigcup_{a\in P} A_a\right| - \sum_{a\in P} c_a$$

for checker-accepted target sets A_a, common target value v and measured cost c_a. For a fresh arm a, the formal insertion law is exact:

$$U(P \cup \{a\}) = U(P) + v\,|A_a \setminus \textstyle\bigcup_{b\in P} A_b| - c_a.$$

No independence model is involved (`portfolioNetValue_insert_eq_add_marginalNetValue`). Comparing a candidate with an independently trained BP arm is therefore exactly a comparison of marginal net values, not standalone yields (`insert_candidate_gt_insert_opportunity_iff`). The positive fixture has a lower-yield PC arm win because its exclusive targets repay equal cost; the negative fixture keeps those targets but raises its cost until the opportunity control wins. This is the theorem-level form of "one more BP seed": diversity is valuable, but not free.

What the search produces is a corpus, and the corpus is now a formal object of its own: a provenance-authenticated many-to-many relation between programs and the sequences they solve, preserving every distinct program witness rather than quotienting programs by what they solve (`provenanceAwareProgramDiscovery_crown`). Its equivalence ladder—exact syntax, finite-prefix agreement, finite-corpus footprint, full extensional equality—is kept honest by a counterexample at each invalid converse, including a finite-probe rigidity witness evaluated through the real synthesis semantics: agreement on a probed prefix does not force extensional equality. The dependence-aware evidence bridge enforces the matching discipline on statistics: witness multiplicity is not independent evidence, exact reuse receives overlap correction, and descendants trained on earlier discoveries do not count as fresh confirmation. Partial agreement can guide search without touching soundness: ranking by first-mismatch depth can never create an acceptance (`partialRanking_cannot_create_acceptance`), and a two-queue scheduler that reserves a baseline fraction bounds how much a misguided heuristic can delay any baseline discovery (`baseline_rank_time_slowdown_bound`)—so semantic shaping is admissible as guidance precisely because it is powerless over acceptance.

12.1 The completed carrier–credit factorial

The architecture argument is now supported by a larger matched experiment, but the experiment does not collapse the axes the theory keeps separate. Four fully frozen generation-eight carriers—recurrent vector, typed workspace, derived-retention belief and learned-retention belief—each receive the same-form zero-output residual adapter. Five update rules begin from identical states and data ledgers within each parent and world; generation ten continues the exact generation-nine adapter and optimizer. Every candidate passes the whole-OEIS checker.

The two-world cross-cell union is 1,617 new entries. By update rule, union/exclusive marginal is prospective PC 968/299, error-coordinate PC 893/66, ordinary BP 869/51, independent BP

866/77, and incremental PC 693/124. By carrier, the same quantity is recurrent 738/396, derived belief 643/281, learned belief 564/220, and workspace 517/183. Prospective PC has the strongest aggregate rule portfolio but does not win every carrier–world cell, and BP remains substantially faster. The result therefore supports the paper's factorization: state carrier and plasticity rule interact, yet each contributes exclusive verified coverage that a scalar ranking would erase.

The checker evidence is itself graded. Shared channels in the original run allowed orphan workers to overwrite support files, so seven fresh single-owner replays rebuilt the terminal evidence chain without retraining. The ordinary and audited numerical summaries agree exactly; the isolated independent verifier is authoritative and passes all 749 checks. Its SHA-256 is `3f08e6c3a57ba3e13a0e387737bbb31cfd990221a7b3b331214b4f9cab47bc7f`. This establishes the reported finite-corpus coverage under the checker, not full-domain correctness of the discovered programs and not efficacy of the still-untested unified carrier.

12.2 A fresh-start architecture–credit relation audit

A separate matched experiment started BP and prospective PC from the same initialization within each of two architectures: the coarse recurrent AC–NMT carrier and the finer Unified carrier. Every proposed program was adjudicated by the same external checker. For endpoint s, define the descriptive interaction

$$I_s = \big(Y_s(\mathrm{Unified}, \mathrm{PC}) - Y_s(\mathrm{Unified}, \mathrm{BP})\big) - \big(Y_s(\mathrm{AC}, \mathrm{PC}) - Y_s(\mathrm{AC}, \mathrm{BP})\big).$$

round and endpoint	AC: PC–BP	Unified: PC–BP	I_s
round 1 verified facts	−355	+8	363
round 1 program identities	−277	+6	283
round 1 targets	−302	+3	305
round 1 fresh program identities	−78	+5	83
round 2 verified facts	−206	+10	216
round 2 program identities	−194	−24	170
round 2 targets	−198	+13	211
round 2 fresh program identities	−78	−8	70

The interaction is endpoint-specific by construction, and the formal development contains an executable witness in which changing only the endpoint projection changes its value (`architectureUpdateInteractionBy`, `architectureInteraction_changes_with_endpoint_projection`). The table therefore does not define one scalar architecture effect. Nor does it identify a causal effect of carrier architecture: architecture is bundled with decoder and graph structure rather than randomized.

Round two exposes the mechanism behind the superficially favorable Unified target contrast. Unified PC covers 95 targets against BP's 82, but represents only 89 program identities against 113, including 71 versus 79 fresh identities. For a relation R between programs and targets, let $d_R(p)$ be the target degree of program p and let $e_R(p)$ be its exclusive contribution, the number of targets supported only by p. The exact deletion identity and bound are

$$|\mathrm{targets}(R \setminus p)| + e_R(p) = |\mathrm{targets}(R)|, \qquad e_R(p) \le d_R(p),$$

formalized as `targetCoverage_withoutProgram_add_exclusive` and `exclusiveTargetContribution_le_programTargetDegree`. A negative fixture shows that equal program degree need not imply equal exclusive contribution (`equal_program_degree_different_exclusive_target_contribution`).

The observed Unified-PC relation contains one previously known program with $d_R = e_R = 36$: one exact same-target reproduction and 35 cross-target reuses. Removing it changes PC target coverage from 95 to 59; the most consequential single BP removal changes 82 to 80. Consequently the ordinary PC–BP target contrast of $+13$ becomes -21 under leave-one-program-out stress. This is evidence for an architecture-conditioned associative reuse channel, but evidence against calling the aggregate advantage robust. Future architecture comparisons should report the authenticated fact relation, program and target projections, fresh identities, program-degree and exclusive-contribution distributions, and the maximum leave-one-program-out change.

12.3 A dependent-proof credit screen

A separate preregistered experiment holds the checked dependent-proof frontier and two-layer pointer-ranker architecture fixed while changing only credit assignment. Five profiles train on the same 1,060-batch ledger, with PC active from update zero. On 604 held-out development problems, ordinary BP solves 286, independent BP 220, error-coordinate PC 244, prospective PC 119, and incremental PC 1. These totals alone would rank BP first. The mechanism trace says something orthogonal: error-coordinate PC remains BP-like (median raw-credit cosine 0.9899), prospective PC is strongly distinct (0.0979), and incremental PC is distinct but does not learn.

Independent-seed replication resolves the useful part of that distinction. Prospective PC reproduces raw-credit cosine 0.0930 and optimizer-displacement cosine 0.3358. Against its matched independent-BP control it solves 160 versus 220 of 600 synthetic held-out problems, yet adds 40 synthetic marginals; it also solves two of four tutorial problems versus zero. On the fixed DTTBench endpoint it proves `False_elim` and `Iff_refl`, while BP proves only `False_elim`. Incremental PC again solves none of the synthetic set and adds no endpoint marginal. Figure 2 shows the replicated mechanism separation: the two statistics are medians over the training run, and optimizer transport pulls prospective PC's nearly orthogonal raw credit (≈ 0.09) partway back toward BP (≈ 0.34) while leaving the incremental–prospective gap intact.

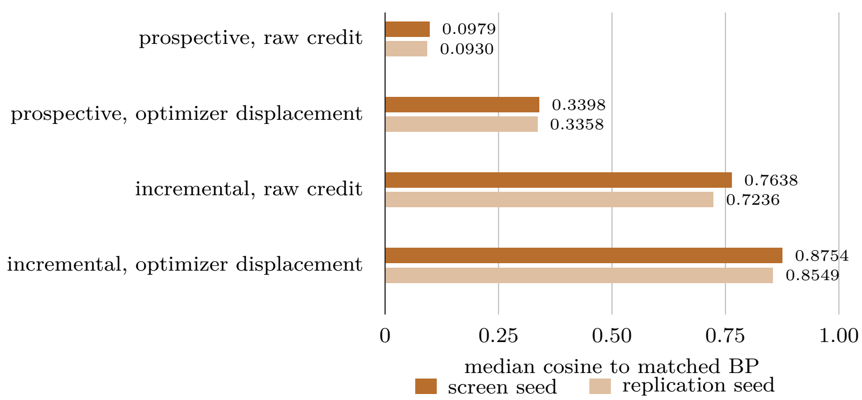

Figure 2: Mechanism replication on the dependent-proof screen. Each pair of bars is one credit statistic on the two independent seeds; agreement is to the second decimal on every row. Prospective PC's raw credit is nearly orthogonal to BP even though the optimizer-transported displacement is not, so raw-credit and displacement cosines must be reported together.

This is the situation the net-value theorem was designed to express: prospective PC is not the highest-coverage arm, but it is not redundant with BP. Whether its marginals repay its additional compute is an empirical input to the exact opportunity-control law. The endpoint had already been opened globally before replica training, so its row is fixed-benchmark replication; the repeated mechanism departure and synthetic held-out marginals carry the cleaner causal interpretation. The experiment uses a shallow pointer ranker, not the unified carrier, and therefore supplies no unified-carrier efficacy claim.

13 What the telemetry can and cannot see

A map is only useful if you can locate yourself on it. Each boundary above is stated in terms of regime parameters—overlap, drift, distortion, cross-slot coupling, commutator energy—and the natural question is whether those are recoverable from what a deployed cell actually logs.

Mostly they are not. Of eleven diagnostics, two are recoverable from primitive telemetry: gain variation from two logged gains, and the scalar spectral envelope from the settling residual. The other nine are *confounded*: for each, two admissible regimes emit identical primitive telemetry and differ in the diagnostic. Each confound comes with the named intervention that resolves it—innovation-residual series for drift, source-identity metadata for overlap, calibration pairs for distortion, paired counterfactual endpoints for the plasticity branch, cross-slot perturbations for coupling, paired operator Jacobians for commutator energy, and Lyapunov measurements for switched stability.

This has a sharp consequence for interpreting experiments. The very quantity that would explain a typed-slot advantage—the cross-slot coupling—is provably not identifiable from ordinary logs (`crossSlotHessian_not_identifiable`), so an observed workspace win cannot be attributed to the mechanism the theory names without adding the perturbation probe. Attribution requires intervention, and the required interventions are enumerated rather than guessed. A search over probe subsets yields exactly two inclusion-minimal suites, differing only in how routed behaviour is identified, and an adaptive procedure that stops at the first recommendation-determining prefix.

14 Scope, and what is assumed rather than proved

We close with the boundary of the boundary theory, stated plainly because several results above are easier than they look.

Typed linearity, now discharged where it can be. The two-timescale results of §11 hold in a model whose effects are linear maps, and that commitment initially carried the argument invisibly: faithfulness reduces to cancellation and reachability to the definition of an image, with no hypothesis to flag. The boundary is now a theorem rather than a hope. Concrete linear attention—outer-product accumulation read by query contraction—provably *instantiates* the model, with additivity over context concatenation derived before anything is packaged as a linear map (`linearAttentionContextEffect_append`); the belief cell's evidence registers instantiate it exactly; and the abstract range condition becomes a concrete rank condition—reachability through an r-dimensional adapter bottleneck holds iff the required update has rank at most r (`rankBudgetReachable_iff_rank_le`), so a rank-one adapter facing a rank-two context effect provably cannot consolidate faithfully. Softmax attention falls outside: normalization couples the context and breaks exact additivity, with a two-token counterexample and a quantitative defect bound (`scalarSoftmaxAttention_not_additive`, `abs_normalizedContextMerge_sub_add_le`). The hand-constructed unnormalized head now has an exact in-context gradient-update theorem. What remains empirical is whether a trained transformer discovers that construction or operates in the near-linear regime—a question on which the evidence is suggestive rather than settled: linear attention is exactly the kernelised recurrence whose state is an accumulated outer product [30]; alternative gradient-step interpretations have been proposed [84]; and empirically a demonstration set's effect can be compressed into a single added vector [85, 86], which is the additive-effect hypothesis in its most directly testable form. These empirical observations are not consequences of the exact construction theorem, and we do not treat them as such.

Derived, not declared, cost models. An earlier draft optimized the consolidation cadence against a posited quadratic trade-off. Deriving the objective from the development's own accounting refuted it—the honest optimum is the corner of §11—and the episode is retained deliberately: a cost model written down rather than derived is an assumption wearing a formula's clothes.

Model families. Except where noted, the results are scalar or matrix linear-Gaussian and quadratic. The depth-repair development reaches full covariance and a nonlinear Jacobian remainder but still proves exact local identities rather than convergence. Trained nonlinear accuracy is an empirical target everywhere it appears.

Negative results are load-bearing. Several of the most useful statements here are refutations—no universal recommendation, stability does not compose, four hypotheses do not suffice for a continual-learning advantage, commuting linear parts do not imply order-independence, nine diagnostics are not identifiable, softmax is not additive. We regard the counterexamples as the more durable half of this work, and the stance has already paid within this note's own lifetime: the first cadence formula did not survive its honest derivation, exactly as anticipated.

Provenance

The development is a Lean 4 library. Definitions, theorem statements and complete proofs are available under the machine-learning tree of the immutable formal snapshot. The unified-carrier development, credit-transport development, utilization boundaries, subsumption partition and two-timescale theory are included in that snapshot. Crown theorems depend only on the standard logical basis (`propext`, `Classical.choice`, `Quot.sound`); the development contains no declared axioms and no unproved placeholders.

References

[1] D. Lopez-Paz and M. Ranzato. Gradient episodic memory for continual learning. *NeurIPS*, 2017; arXiv:1706.08840.

[2] M. Farajtabar, N. Azizan, A. Mott, and A. Li. Orthogonal gradient descent for continual learning. *AISTATS*, PMLR 108, 2020; arXiv:1910.07104.

[3] G. Saha, I. Garg, and K. Roy. Gradient projection memory for continual learning. *ICLR*, 2021; arXiv:2103.09762.

[4] S. Lin, L. Yang, D. Fan, and J. Zhang. TRGP: Trust region gradient projection for continual learning. *ICLR*, 2022; arXiv:2202.02931.

[5] S. I. Mirzadeh, A. Chaudhry, D. Yin, H. Hu, R. Pascanu, D. Gorur, and M. Farajtabar. Wide neural networks forget less catastrophically. *ICML*, PMLR 162, 2022; arXiv:2110.11526.

[6] E. Guha and V. Lakshman. On the diminishing returns of width for continual learning. *ICML*, PMLR 235, 2024; arXiv:2403.06398.

[7] J. Graldi, A. Breccia, G. Lanzillotta, T. Hofmann, and L. Noci. The importance of being lazy: scaling limits of continual learning. *ICML*, PMLR 267, 2025; arXiv:2506.16884.

[8] G. M. van de Ven. On the computation of the Fisher information in continual learning. *ICLR Blogpost*, 2025; arXiv:2502.11756.

[9] T.-C. Kao, K. T. Jensen, G. M. van de Ven, A. Bernacchia, and G. Hennequin. Natural continual learning: Success is a journey, not (just) a destination. *NeurIPS*, 2021; arXiv:2106.08085.

[10] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. S. Torr. Riemannian Walk for incremental learning: Understanding forgetting and intransigence. *ECCV*, 2018; arXiv:1801.10112.

[11] C. V. Nguyen, Y. Li, T. D. Bui, and R. E. Turner. Variational continual learning. *ICLR*, 2018; arXiv:1710.10628.

[12] C. Zeno, I. Golan, E. Hoffer, and D. Soudry. Task agnostic continual learning using online variational Bayes. arXiv:1803.10123, 2018.

[13] J. Knoblauch, H. Husain, and T. Diethe. Optimal continual learning has perfect memory and is NP-hard. *ICML*, PMLR 108, 2020; arXiv:2006.05188.

[14] R. Aljundi, M. Lin, B. Goujaud, and Y. Bengio. Gradient based sample selection for online continual learning. *NeurIPS*, 2019; arXiv:1903.08671.

[15] H. Shin, J. K. Lee, J. Kim, and J. Kim. Continual learning with deep generative replay. *NeurIPS*, 2017; arXiv:1705.08690.

[16] P. Buzzega, M. Boschini, A. Porrello, D. Abati, and S. Calderara. Dark experience for general continual learning: a strong, simple baseline. *NeurIPS*, 2020; arXiv:2004.07211.

[17] S. I. Mirzadeh, M. Farajtabar, D. Gorur, R. Pascanu, and H. Ghasemzadeh. Linear mode connectivity in multitask and continual learning. *ICLR*, 2021; arXiv:2010.04495.

[18] R. Aljundi, K. Kelchtermans, and T. Tuytelaars. Task-free continual learning. *CVPR*, 2019; arXiv:1812.03596.

[19] A. Banayeeanzade, M. Soltanolkotabi, and M. Rostami. Theoretical insights into overparameterized models in multi-task and replay-based continual learning. *Transactions on Machine Learning Research*, 2025; arXiv:2408.16939.

[20] N. Y. Masse, G. D. Grant, and D. J. Freedman. Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization. *Proceedings of the National Academy of Sciences*, 115(44):E10467–E10475, 2018; DOI:10.1073/pnas.1803839115; arXiv:1802.01569.

[21] L. Pellegrini, G. Graffieti, V. Lomonaco, and D. Maltoni. Latent replay for real-time continual learning. *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 10203–10209, 2020; DOI:10.1109/IROS45743.2020.9341460; arXiv:1912.01100.

[22] M. Seo, H. Koh, and J. Choi. Budgeted online continual learning by adaptive layer freezing and frequency-based sampling. *ICLR*, 2025; arXiv:2410.15143.

[23] A. Prabhu, H. A. A. K. Hammoud, P. Dokania, P. H. S. Torr, S.-N. Lim, B. Ghanem, and A. Bibi. Computationally budgeted continual learning: what does matter? *CVPR*, 2023; arXiv:2303.11165.

[24] H. Hemati et al. Class-incremental learning with repetition. *Conference on Lifelong Learning Agents*, 2023; arXiv:2301.11396.

[25] Y. Ghunaim et al. Real-time evaluation in online continual learning: A new hope. *CVPR*, 2023; arXiv:2302.01047.

[26] A. Prabhu, Z. Cai, P. Dokania, P. H. S. Torr, V. Koltun, and O. Sener. Online continual learning without the storage constraint. *ICML*, 2023; arXiv:2305.09253.

[27] Y. Sun et al. Learning to (learn at test time): RNNs with expressive hidden states. arXiv:2407.04620, 2024.

[28] B. Liu et al. Longhorn: State space models are amortized online learners. arXiv:2407.14207, 2024.

[29] S. Yang et al. Gated delta networks: Improving Mamba2 with the delta rule. arXiv:2412.06464, 2024.

[30] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. *ICML*, 2020; arXiv:2006.16236.

[31] J. von Oswald et al. Transformers learn in-context by gradient descent. *ICML*, 2023; arXiv:2212.07677.

[32] L. Ninite and R. M. Jungers. Iterative graph lifting for automatic design of path-complete stability certificates. arXiv:2607.00637, 2026.

[33] P. Deidda, N. Guglielmi, and F. Tudisco. Nonlinear joint spectral radius. arXiv:2507.11314, 2025.

[34] P. Ashwin and C. Postlethwaite. Designing heteroclinic and excitable networks in phase space using two populations of coupled cells. arXiv:1506.03212, 2015.

[35] A. Voit and H. Meyer-Ortmanns. Hierarchical heteroclinic networks: controlling the time evolution of complex systems. arXiv:1806.11039, 2018.

[36] M. R. Zhang, J. Lucas, G. Hinton, and J. Ba. Lookahead optimizer: k steps forward, 1 step back. arXiv:1907.08610, 2019.

[37] Q. Pham, C. Liu, and S. C. H. Hoi. DualNet: continual learning, fast and slow. arXiv:2110.00175, 2021.

[38] A. Prabhu, S. Sinha, P. Kumaraguru, P. H. S. Torr, O. Sener, and P. K. Dokania. Random representations outperform online continually learned representations. *NeurIPS*, 2024; arXiv:2402.08823.

[39] T. L. Hayes and C. Kanan. Lifelong machine learning with deep streaming linear discriminant analysis. *CVPR Workshops*, 2020; arXiv:1909.01520.

[40] D. Goswami, Y. Liu, B. Twardowski, and J. van de Weijer. FeCAM: exploiting the heterogeneity of class distributions in exemplar-free continual learning. *NeurIPS*, 2023; arXiv:2309.14062.

[41] M. K. Nori and I.-M. Kim. Task confusion and catastrophic forgetting in class-incremental learning: a mathematical framework for discriminative and generative modelings. arXiv:2410.20768, 2024.

[42] M. D. McDonnell, D. Gong, A. Parveneh, E. Abbasnejad, and A. van den Hengel. RanPAC: random projections and pre-trained models for continual learning. *NeurIPS*, 2023; arXiv:2307.02251.

[43] H. Zhuang, Z. Weng, H. Wei, R. Xie, K.-A. Toh, and Z. Lin. ACIL: analytic class-incremental learning with absolute memorization and privacy protection. arXiv:2205.14922, 2022.

[44] H. Zhuang, Y. Liu, R. He, K. Tong, Z. Zeng, C. Chen, Y. Wang, and L.-P. Chau. F-OAL: forward-only online analytic learning with fast training and low memory footprint in class-incremental learning. arXiv:2403.15751, 2024.

[45] L. Pinchetti, C. Qi, O. Lokshyn, G. Oliviers, C. Emde, M. Tang, A. M'Charrak, S. Frieder, B. Menzat, R. Bogacz, T. Lukasiewicz, and T. Salvatori. Benchmarking predictive coding networks—made simple. *ICLR*, 2025; arXiv:2407.01163.

[46] L. Peng, J. Elenter, J. Agterberg, A. Ribeiro, and R. Vidal. LoRanPAC: low-rank random features and pre-trained models for bridging theory and practice in continual learning. *ICLR*, 2025; arXiv:2410.00645.

[47] Q. Cheng, Y. Wan, L. Wu, C. Hou, and L. Zhang. Continuous subspace optimization for continual learning. arXiv:2505.11816, 2025.

[48] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz. Invariant risk minimization. arXiv:1907.02893, 2019.

[49] E. Rosenfeld, P. Ravikumar, and A. Risteski. The risks of invariant risk minimization. *ICLR*, 2021; arXiv:2010.05761.

[50] G. Ortiz-Jimenez, A. Favero, and P. Frossard. Task arithmetic in the tangent space: improved editing of pre-trained models. *NeurIPS*, 2023; arXiv:2305.12827.

[51] M. Wortsman, V. Ramanujan, R. Liu, A. Kembhavi, M. Rastegari, J. Yosinski, and A. Farhadi. Supermasks in superposition. *NeurIPS*, 2020; arXiv:2006.14769.

[52] Z. Wang, Z. Zhang, C.-Y. Lee, H. Zhang, R. Sun, X. Ren, G. Su, V. Perot, J. Dy, and T. Pfister. Learning to prompt for continual learning. *CVPR*, 2022; arXiv:2112.08654.

[53] Z. Wang, Z. Zhang, S. Ebrahimi, R. Sun, H. Zhang, C.-Y. Lee, X. Ren, G. Su, V. Perot, J. Dy, and T. Pfister. DualPrompt: complementary prompting for rehearsal-free continual learning. *ECCV*, 2022; arXiv:2204.04799.

[54] J. S. Smith, L. Karlinsky, V. Gutta, P. Cascante-Bonilla, D. Kim, A. Arbelle, R. Panda, R. Feris, and Z. Kira. CODA-Prompt: continual decomposed attention-based prompting for rehearsal-free continual learning. *CVPR*, 2023; arXiv:2211.13218.

[55] L. Wang, J. Xie, X. Zhang, M. Huang, H. Su, and J. Zhu. Hierarchical decomposition of prompt-based continual learning: rethinking obscured sub-optimality. arXiv:2310.07234, 2023.

[56] M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, and L. Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. *ICML*, PMLR 162, 2022; arXiv:2203.05482.

[57] S. Blanes, F. Casas, and A. Murua. Splitting and composition methods in the numerical integration of differential equations. arXiv:0812.0377, 2008.

[58] G. Hinton. The Forward-Forward Algorithm: Some Preliminary Investigations. arXiv:2212.13345, 2022.

[59] S. S. Schoenholz, J. Gilmer, S. Ganguli, and J. Sohl-Dickstein. Deep information propagation. *ICLR*, 2017; arXiv:1611.01232.

[60] B. Poole, S. Lahiri, M. Raghu, J. Sohl-Dickstein, and S. Ganguli. Exponential expressivity in deep neural networks through transient chaos. arXiv:1606.05340, 2016.

[61] L. Xiao, Y. Bahri, J. Sohl-Dickstein, S. S. Schoenholz, and J. Pennington. Dynamical isometry and a mean field theory of CNNs: how to train 10,000-layer vanilla convolutional neural networks. *ICML*, 2018; arXiv:1806.05393.

[62] G. Yang et al. Tensor programs V: tuning large neural networks via zero-shot hyperparameter transfer. arXiv:2203.03466, 2022.

[63] B. Bordelon, L. Noci, M. B. Li, B. Hanin, and C. Pehlevan. Depthwise hyperparameter transfer in residual networks: dynamics and scaling limit. arXiv:2309.16620, 2023.

[64] G. Yang, D. Yu, C. Zhu, and S. Hayou. Tensor programs VI: feature learning in infinite-depth neural networks. arXiv:2310.02244, 2023.

[65] S. Santurkar, D. Tsipras, A. Ilyas, and A. Madry. How does batch normalization help optimization? *NeurIPS*, 2018; arXiv:1805.11604.

[66] S. Löwe, P. O'Connor, and B. S. Veeling. Putting an end to end-to-end: gradient-isolated learning of representations. *NeurIPS*, 2019; arXiv:1905.11786.

[67] M. Riemer et al. Learning to learn without forgetting by maximizing transfer and minimizing interference. *ICLR*, 2019; arXiv:1810.11910.

[68] Y. Bengio, T. Deleu, N. Rahaman, R. Ke, S. Lachapelle, O. Bilaniuk, A. Goyal, and C. Pal. A meta-transfer objective for learning to disentangle causal mechanisms. arXiv:1901.10912, 2019.

[69] R. Perry, J. von Kügelgen, and B. Schölkopf. Causal discovery in heterogeneous environments under the sparse mechanism shift hypothesis. *NeurIPS*, 2022; arXiv:2206.02013.

[70] M. Sensoy, L. Kaplan, and M. Kandemir. Evidential deep learning to quantify classification uncertainty. *NeurIPS*, 2018; arXiv:1806.01768.

[71] A. Malinin and M. Gales. Predictive uncertainty estimation via Prior Networks. *NeurIPS*, 2018; arXiv:1802.10501.

[72] S. Lee, J. Ha, D. Zhang, and G. Kim. A neural Dirichlet process mixture model for task-free continual learning. *ICLR*, 2020; arXiv:2001.00689.

[73] S. Dohare, J. F. Hernandez-Garcia, P. Rahman, A. R. Mahmood, and R. S. Sutton. Maintaining plasticity in deep continual learning. arXiv:2306.13812, 2024.

[74] L. Frati, N. Traft, J. Clune, and N. Cheney. Reset it and forget it: relearning last-layer weights improves continual and transfer learning. arXiv:2310.07996, 2024.

[75] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network representations revisited. *ICML*, 2019; arXiv:1905.00414.

[76] M. Davari, S. Horoi, A. Natik, G. Lajoie, G. Wolf, and E. Belilovsky. Reliability of CKA as a similarity measure in deep learning. arXiv:2210.16156, 2022.

[77] L. N. Smith. Cyclical learning rates for training neural networks. *WACV*, 2017; arXiv:1506.01186.

[78] L. N. Smith and N. Topin. Super-convergence: very fast training of neural networks using large learning rates. arXiv:1708.07120, 2017.

[79] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *ICLR*, 2021; arXiv:2103.00065.

[80] A. Damian, E. Nichani, and J. D. Lee. Self-stabilization: the implicit bias of gradient descent at the edge of stability. *ICLR*, 2023; arXiv:2209.15594.

[81] J. M. Cohen, A. Damian, A. Talwalkar, J. Z. Kolter, and J. D. Lee. Understanding optimization in deep learning with central flows. *ICLR*, 2025; arXiv:2410.24206.

[82] B. O'Donoghue and E. Candès. Adaptive restart for accelerated gradient schemes. *Foundations of Computational Mathematics*, 15:715–732, 2015; arXiv:1204.3982.

[83] P. Izmailov, D. Podoprikhin, T. Garipov, D. Vetrov, and A. G. Wilson. Averaging weights leads to wider optima and better generalization. *UAI*, 2018; arXiv:1803.05407.

[84] D. Dai et al. Why can GPT learn in-context? Language models implicitly perform gradient descent as meta-optimizers. arXiv:2212.10559, 2022.

[85] R. Hendel, M. Geva, and A. Globerson. In-context learning creates task vectors. *EMNLP Findings*, 2023; arXiv:2310.15916.

[86] E. Todd et al. Function vectors in large language models. arXiv:2310.15213, 2023.